

# Agentic AI: Connecting Perception, Reasoning, and Action in Smart Mobility and Logistics

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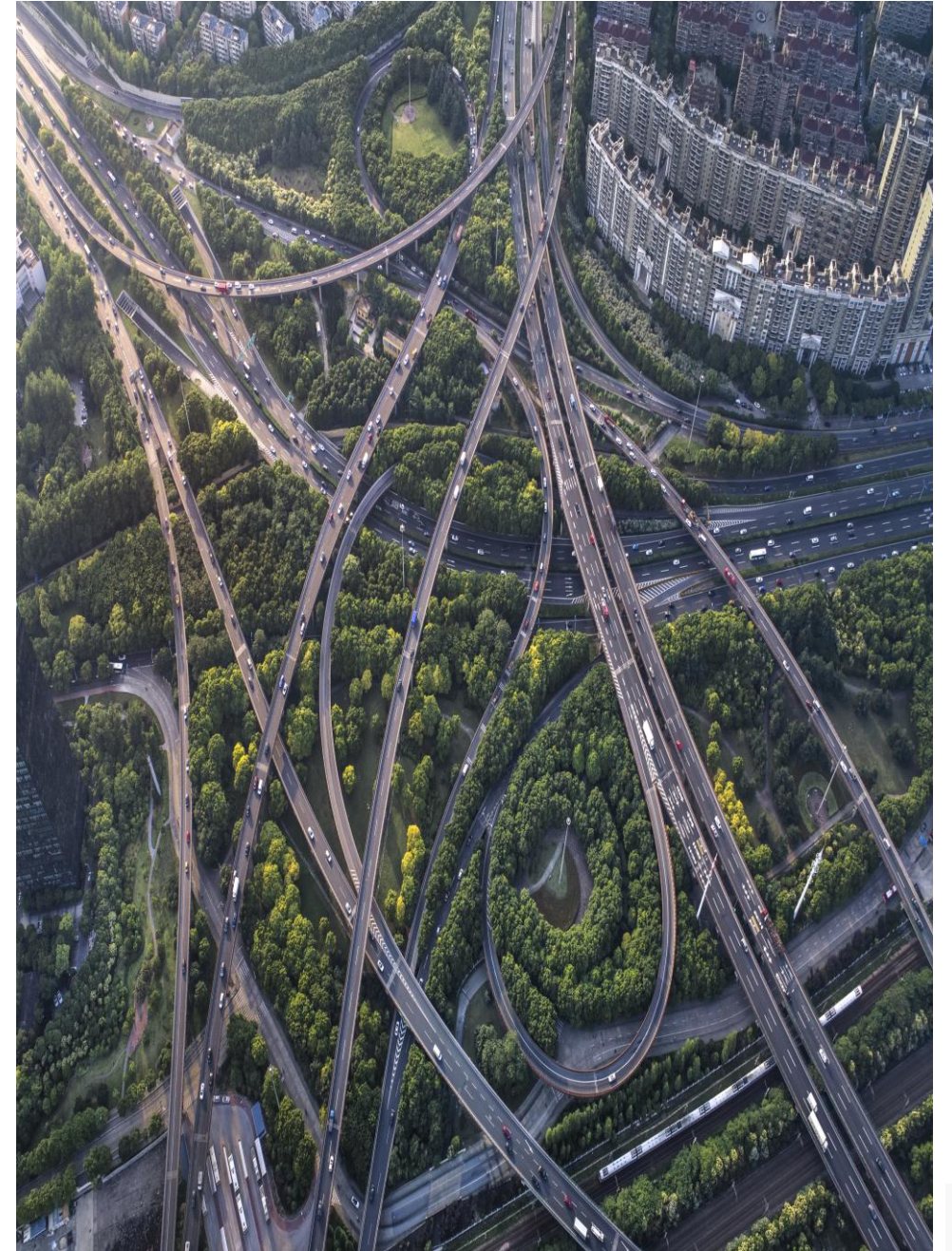
Director of AI4SM Lab at IRC SML

IEEE ITSS – Saudi Chapter Chair



# Outline

- » About AI for Smart Mobility Lab
- » Motivating Scenarios
- » What is Agentic AI?
- » AI Agent Components
- » Use Cases in Smart Mobility and Logistics



# AI for Smart Mobility Lab at KFUPM

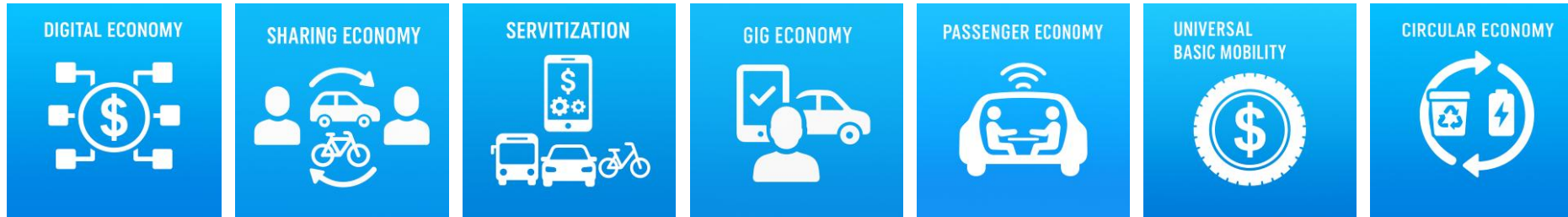


# AI for Smart Mobility Lab: Mission

- » Our mission is to advance **smart mobility** as a transformative enabler of **sustainable development**.
- » Our research focuses on the **intersection of AI and mobility systems, services and business models**.



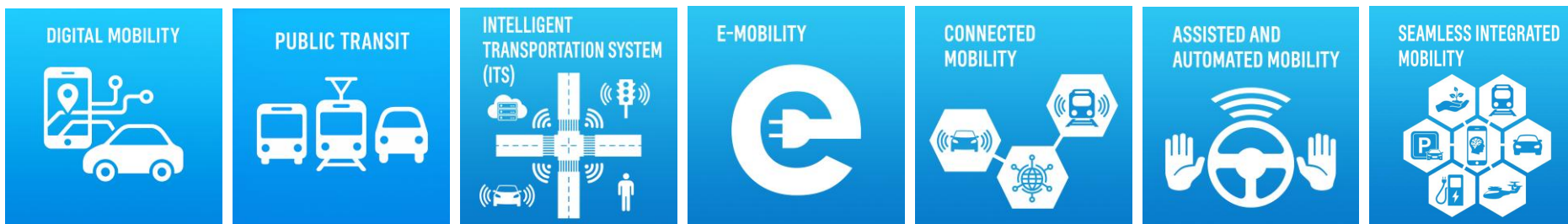
# AI for Smart Mobility Lab: Smart Mobility



Existing and emerging smart mobility business models



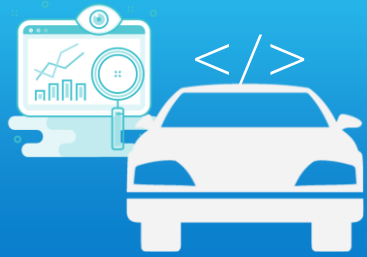
Existing and emerging smart mobility services



Existing and emerging smart mobility systems

# AI for Smart Mobility Lab: Ongoing Projects

## SDV CONTEXTUAL OBSERVABILITY



- **Title:** Contextual Observability of Software-Defined Vehicles
- **Objective:** Develop a testbed for software-defined vehicle (SDV) contextual observability.
- **Collaboration:** IRC SML and auto OEMs, NGOs and Suppliers

## SEAMLESS INTEGRATED MOBILITY



- **Title:** Agentic AI-based Framework for SIM
- **Objective:** Develop as a unified platform that integrates multimodal transportation options.
- **Collaboration:** RCRC, MIT, VTTI

## LAST MILE DELIVERY



- **Title:** SmartDispatch: AI-driven Optimization for Eco-Efficient Last-Mile Delivery
- **Objective:** Develop an AI-driven routing model for eco-efficient last-mile delivery.
- **Collaboration:** IRC SML KFUPM



- **Title:** Enabling Cybersecurity Adaptation in Software Architecture
- **Objective:** What-if architecture analysis of existing software systems with poor or no records of architectural decisions.
- **Collaboration:** UAB, Chile

# Agentic AI: Motivating Scenarios



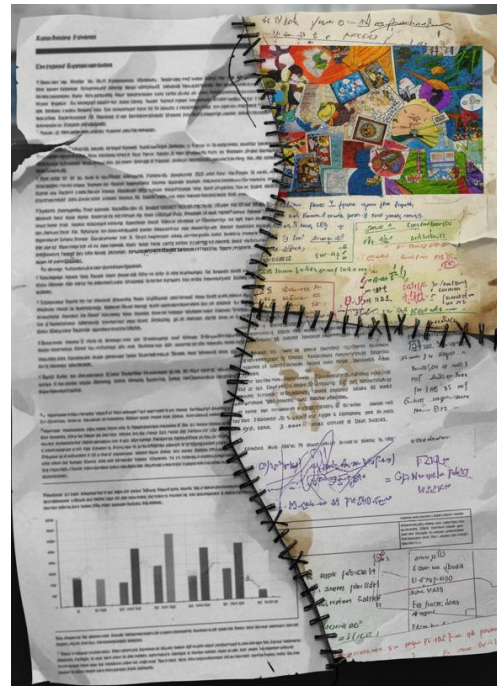
# Motivating Scenarios



Researcher

User query: Write a systematic review paper about smart mobility

Response: Frankenstein Paper



**REJECTED**



## ChatGPT

### Examples

"Explain quantum computing in simple terms" →

"Got any creative ideas for a 10 year old's birthday?" →

"How do I make an HTTP request in Javascript?" →

### Capabilities

Remembers what user said earlier in the conversation

Allows user to provide follow-up corrections

Trained to decline inappropriate requests

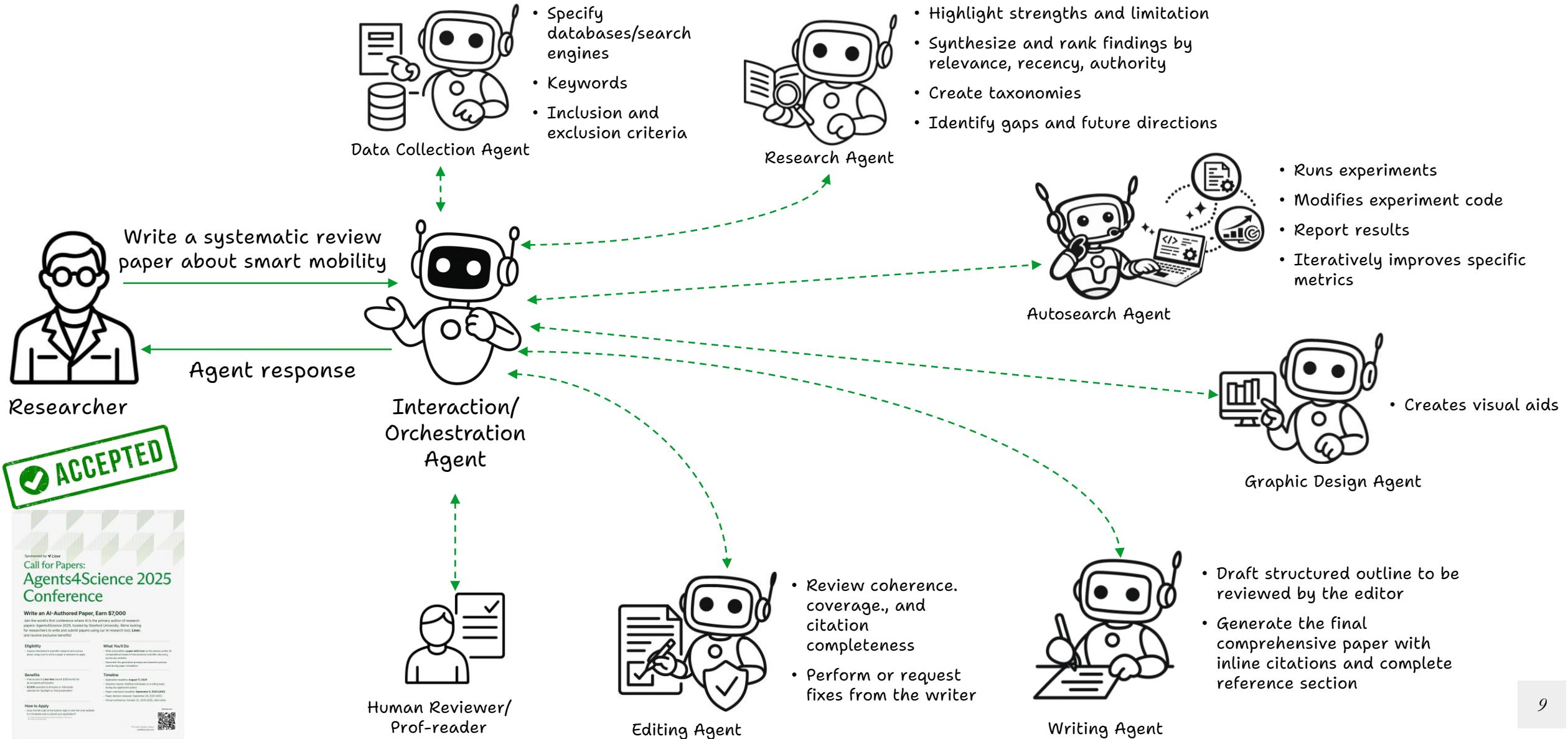
### Limitations

May occasionally generate incorrect information

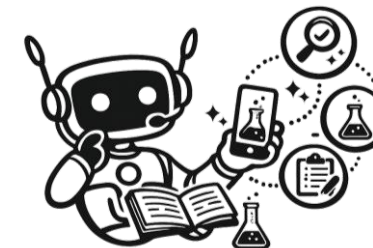
May occasionally produce harmful instructions or biased content

Limited knowledge of world and events after 2021

# Motivating Scenarios



# Motivating Scenarios



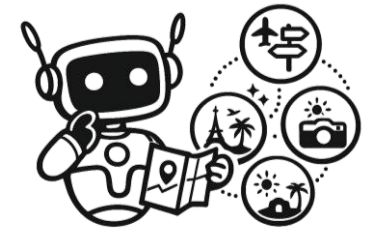
Research Assistant

- » **Researcher Agent** — gather evidence across papers, web, repos, docs
- » **Reviewer Agent** — simulated peer review with severity-graded feedback
- » **Writer Agent** — structured drafts from research notes
- » **Verifier Agent** — inline citations, source URL verification, dead link cleanup

<https://github.com/getcompanion-ai/feynman>

# Motivating Scenarios

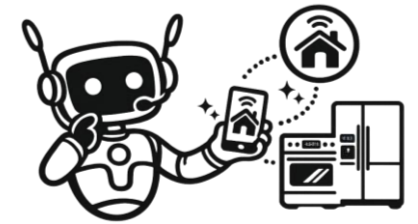
- Got an Intent-driven AI Assistant?



Tour Guide

# Motivating Scenarios

- Got an Intent-driven AI Assistant?



Home Physical AI Assistant



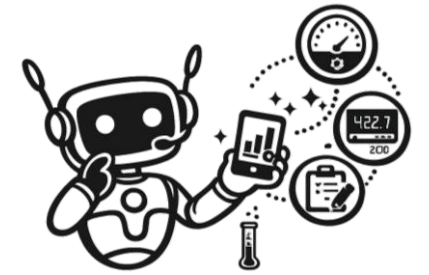
**nVIDIA®**



Figure Helix: <https://www.youtube.com/watch?v=Z3yQHYNXPws>

# Motivating Scenarios

- Got an Intent-driven AI Assistant?

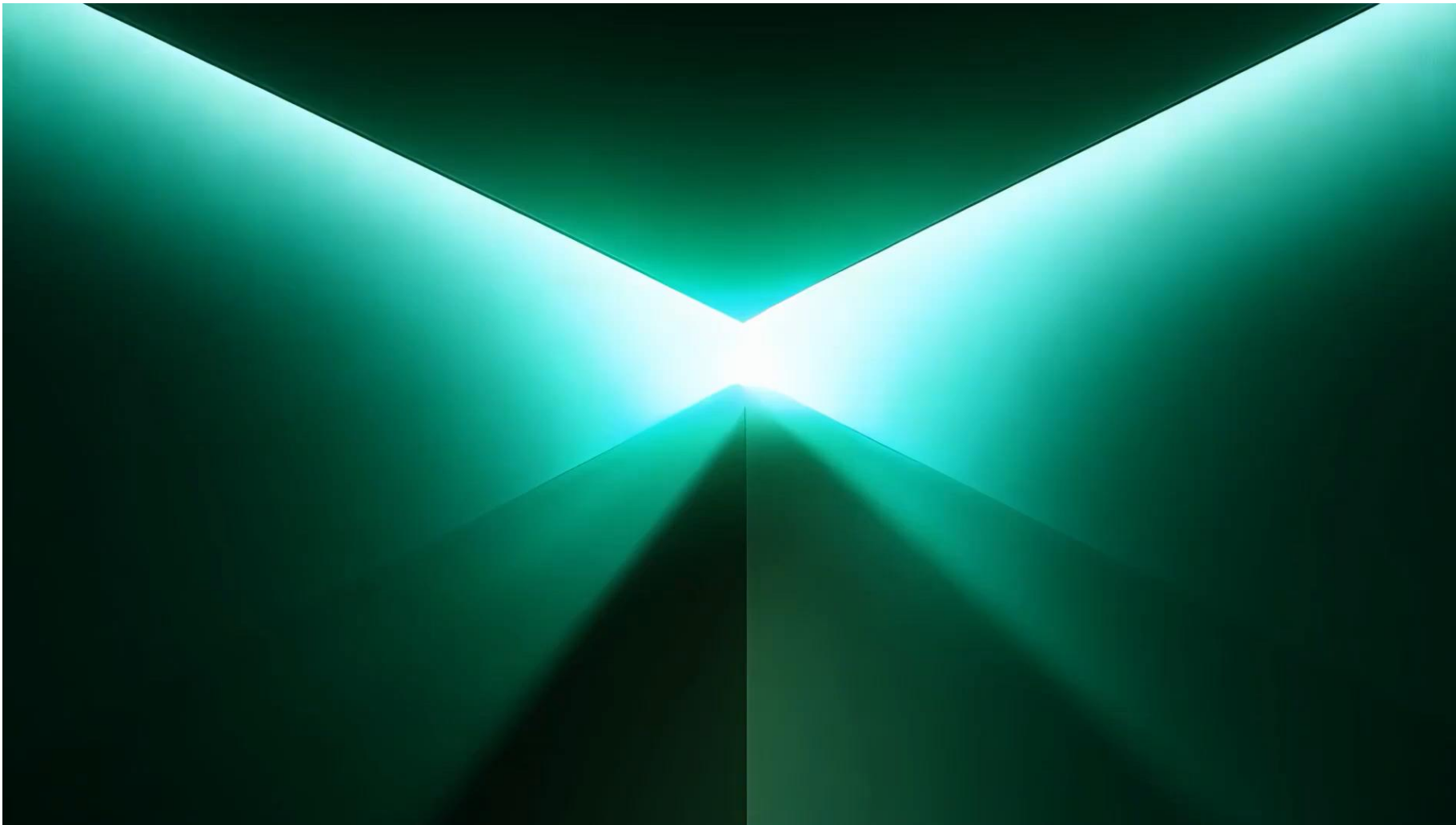


Inspection Assistant

Credit: Gemini Robotics-ER 1.6, Google and Boston Dynamics: <https://www.youtube.com/watch?v=kBwxmII2yHQ>

# Motivating Scenarios

- Got an Intent-driven AI Assistant?

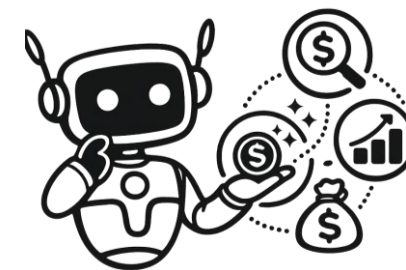
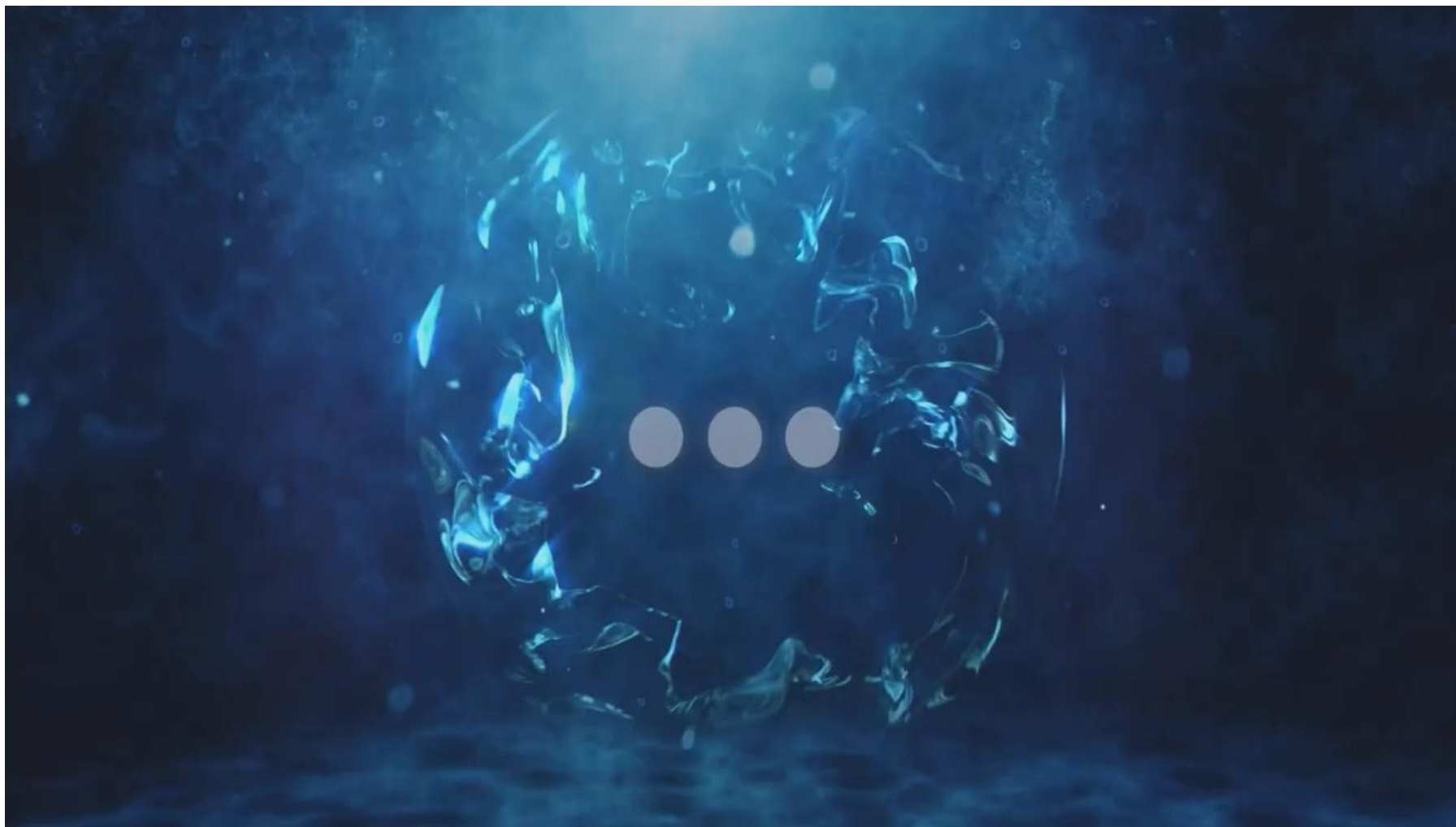


- » Task Manager
- » Meeting Manager
- » File Manager
- » Email Manager
- » Leave Manager
- » Expense Manager
- » Business Trip Manager
- » Policy Manager
- » Job Requisition Manager
- » Applicant Tracking System
- » Interview Manager
- » Job Offer Manager
- » Payroll Manager
- » Onboarding Manager
- » Offboarding Manager
- » Performance Manager
- » Career Website
- » NDA Manager
- » Project Manager

Humain: <https://www.youtube.com/watch?v=UBQWwE4BLqg>

# Motivating Scenarios

- Got an Intent-driven AI Assistant?



Professional Financial Assistant

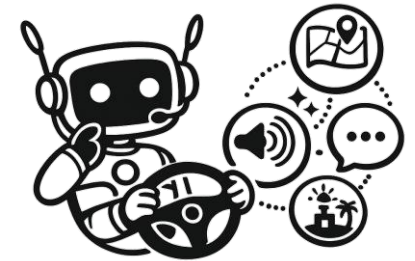
The Family Office Wealth Mermaid: <https://www.youtube.com/watch?v=57tZekT427w>

# Motivating Scenarios

- Got an Intent-driven AI Assistant?



Credit: Jimmy Fallon



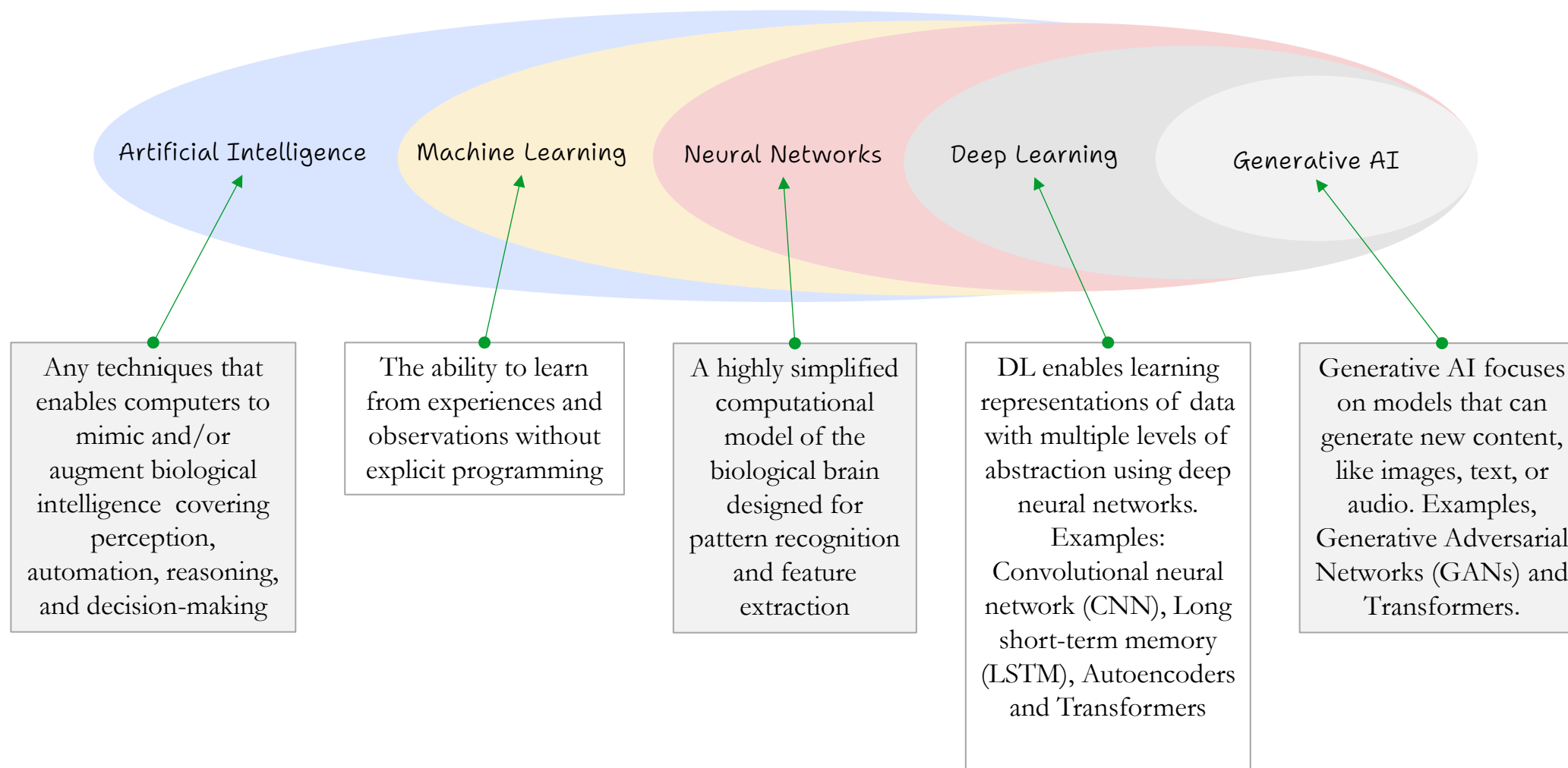
In-vehicle Assistant



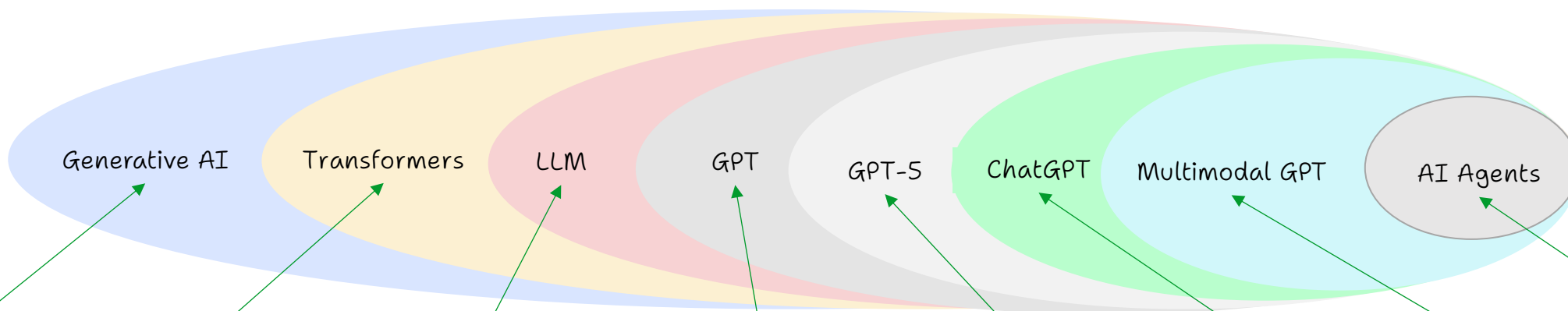
# What is *Agentic* AI?



# The Alphabet Soup of AI



# The Alphabet Soup of AI



Generative AI focuses on models that can generate new content, like images, text, or audio. Examples, Generative Adversarial Networks (GANs) and Transformers.

An example of a technology used in generative AI. It is a revolutionary deep learning architecture introduced by Google in 2017 that allows models to understand and generate language efficiently.

Large Language Models (LLMs) focus on processing and generating human-like text. Examples: Bidirectional Encoder Representations from Transformers (BERT) and Generative Pre-trained Transformer (GPT).

Generative Pre-trained Transformers (GPT) is a specific subset of Generative AI that uses transformers for text generation.

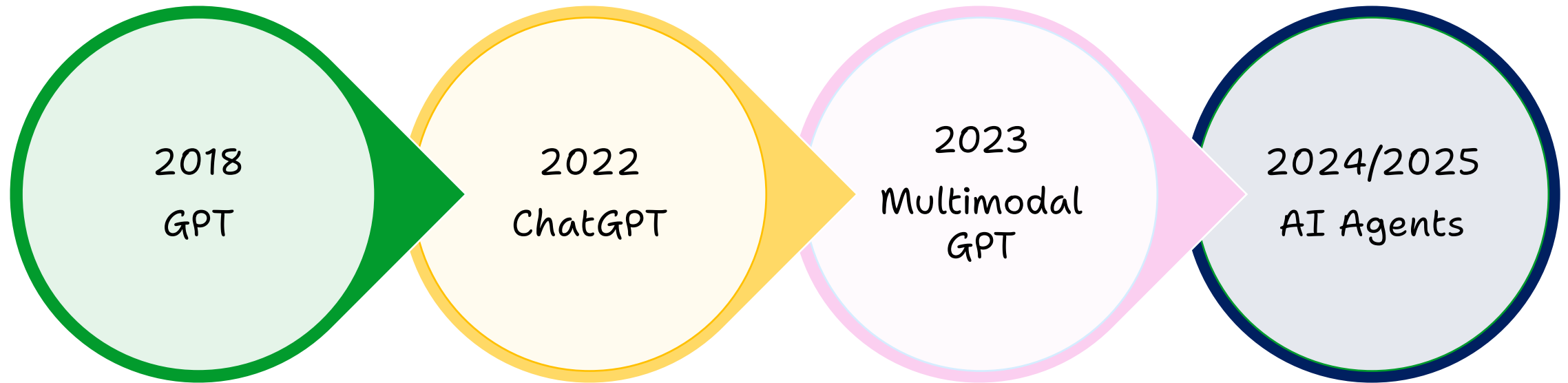
An advanced LLMs, built on transformer architecture, trained on vast datasets to generate human-like responses.

A specific application of GPT-5, optimized for conversational AI and interactive use.

A version of the GPT model, like GPT-4o, that processes and generates both text and visual inputs, functioning as a Vision Large Model (VLM).

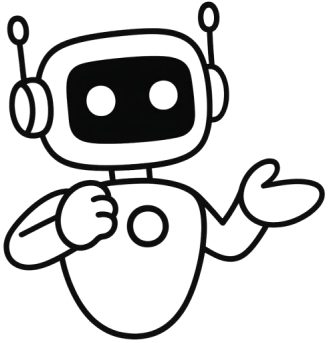
Intelligent connection between reasoning and action (AI: Augmented Intelligence)

# The Alphabet Soup of AI



# What is AI Agent?

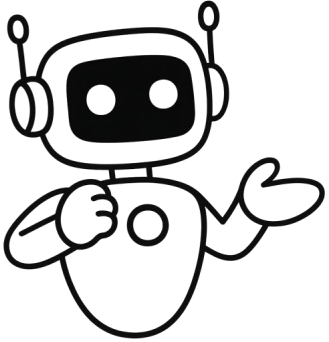
## » LLM/LRM-based AI Agents



An AI agent is a **goal-oriented autonomous computational entity** that **connects reasoning and action** using large language models (LLMs) or Large Reasoning Models (LRMs), memory systems, and external tools to **achieve contextually intelligent outcomes**.

# What is AI Agent?

## » LLM/LRM-based AI Agents

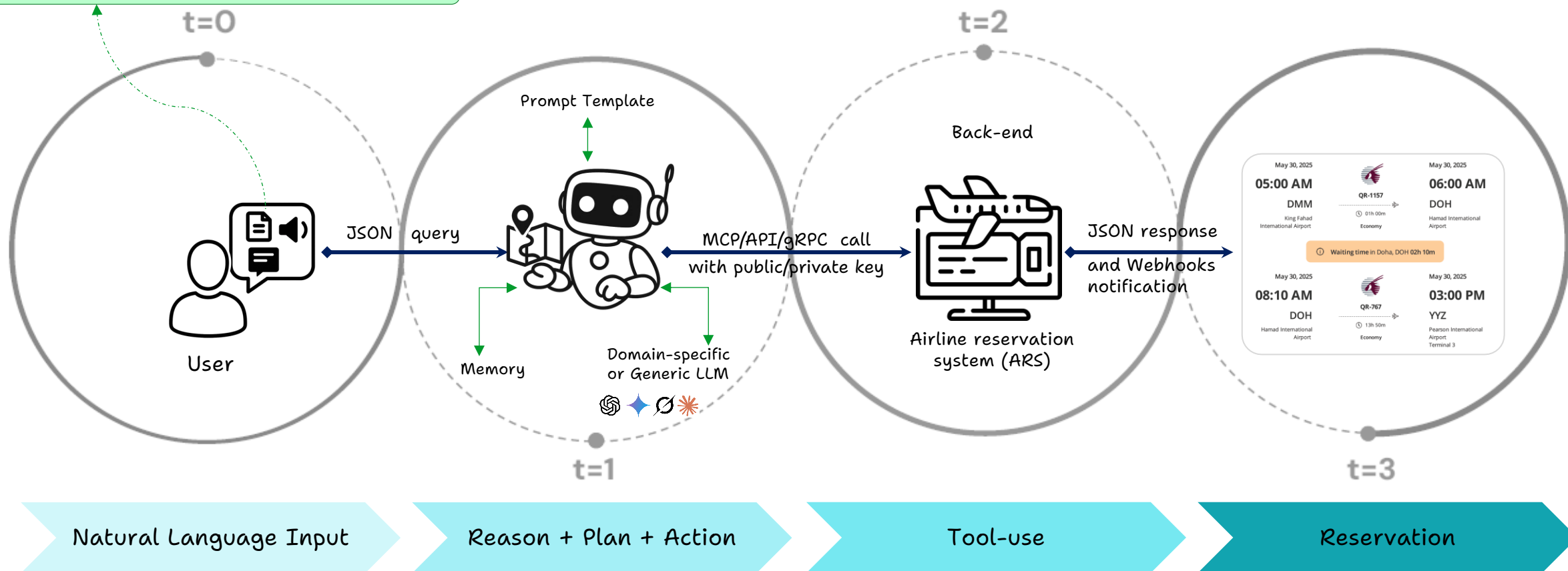


1. Goal-oriented
2. Autonomous
3. Connects reasoning and action
4. Achieve contextually intelligent outcomes.

# What is AI Agent?

## » Intent-driven Agentic Workflow

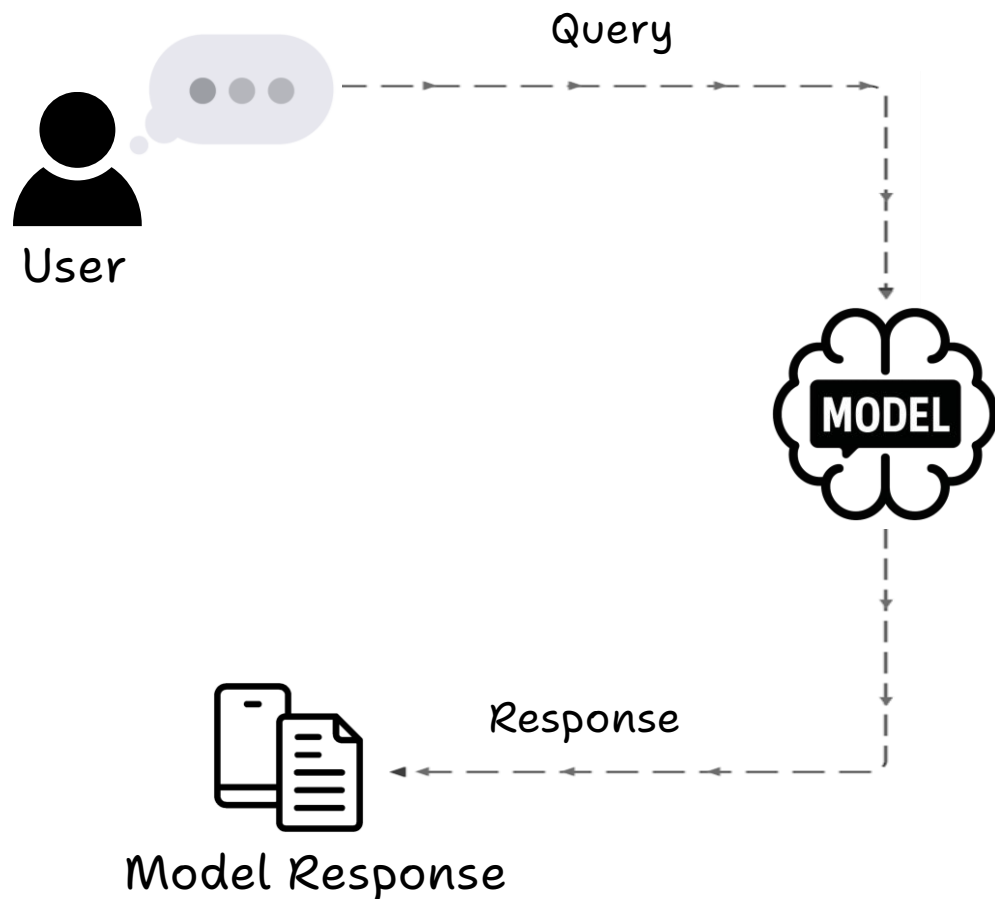
Could you please help me book a flight from Dammam to Toronto on May 30? I'd prefer a flight with no more than one stop, and I'd like to keep the total travel time as reasonable as possible. If there are multiple options, please prioritize the one with the shortest layover and a departure time after 5:00 AM local time in Dammam and arrival before 5:00PM local time in Toronto.



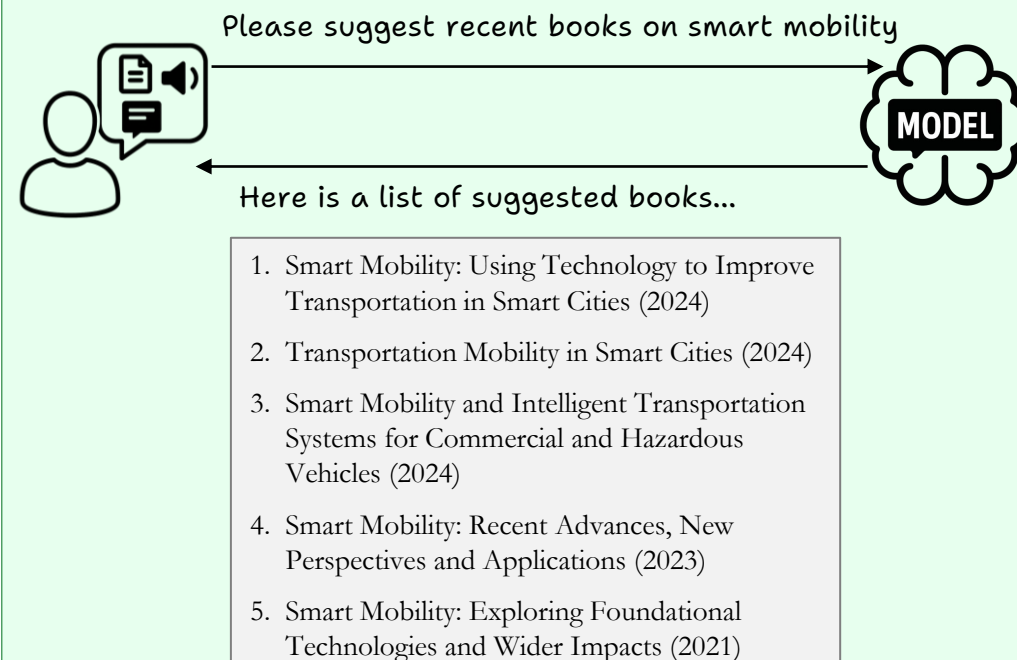
# What is AI Agent?

## » Design Patterns: Basic Responder

Agency Level

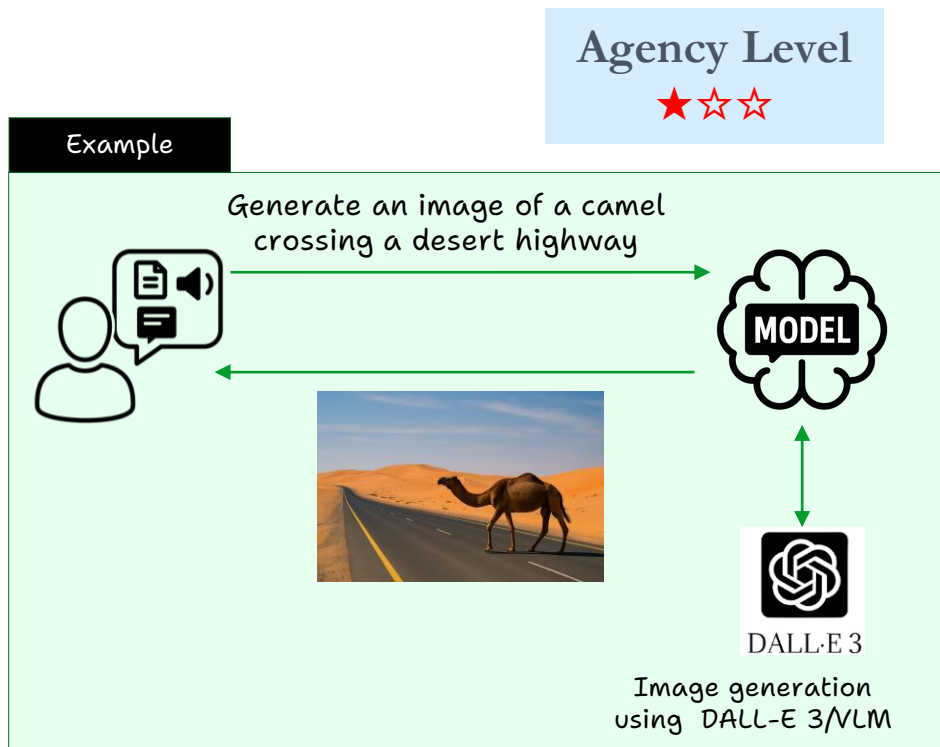
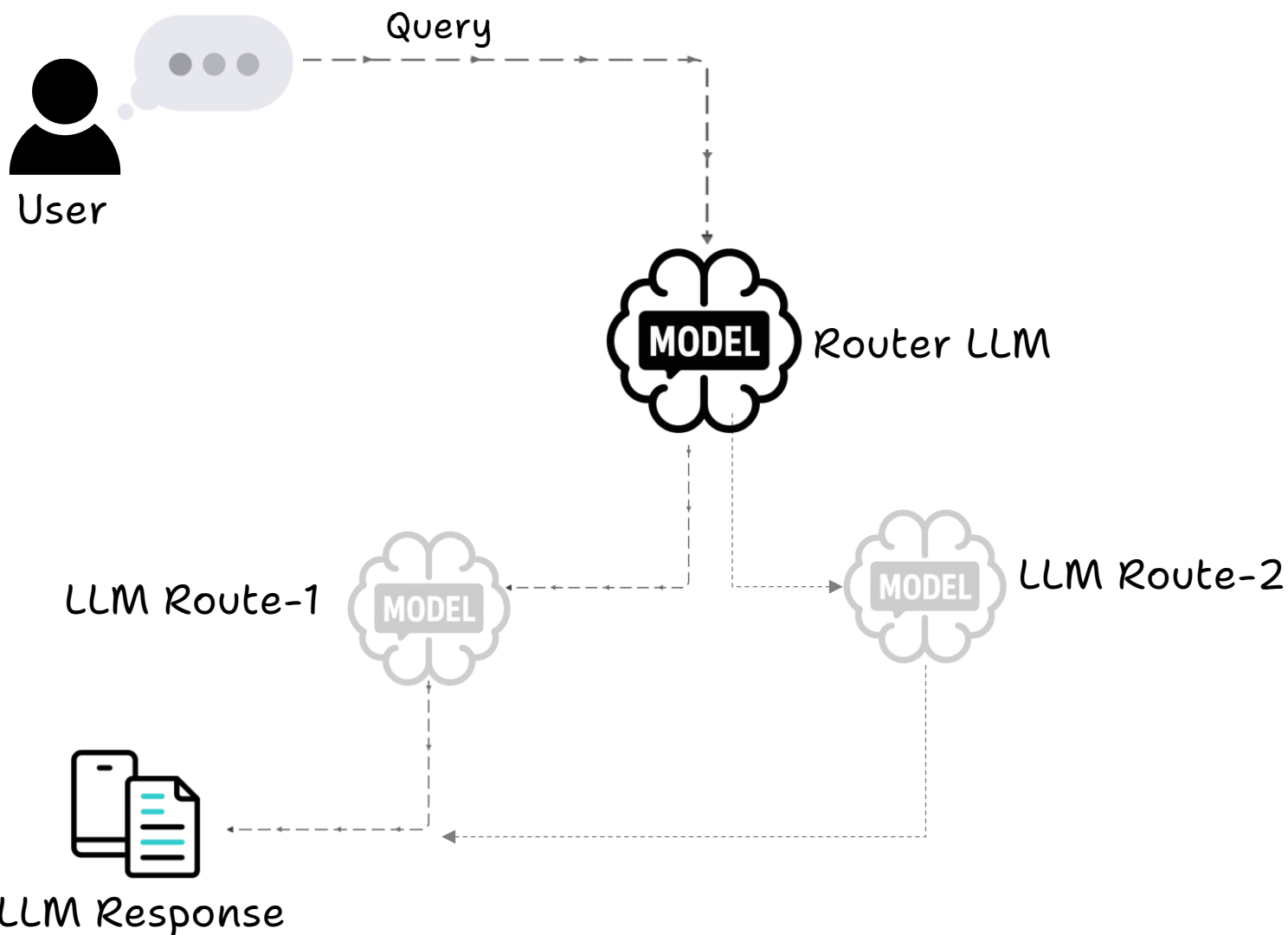


### Example



# What is AI Agent?

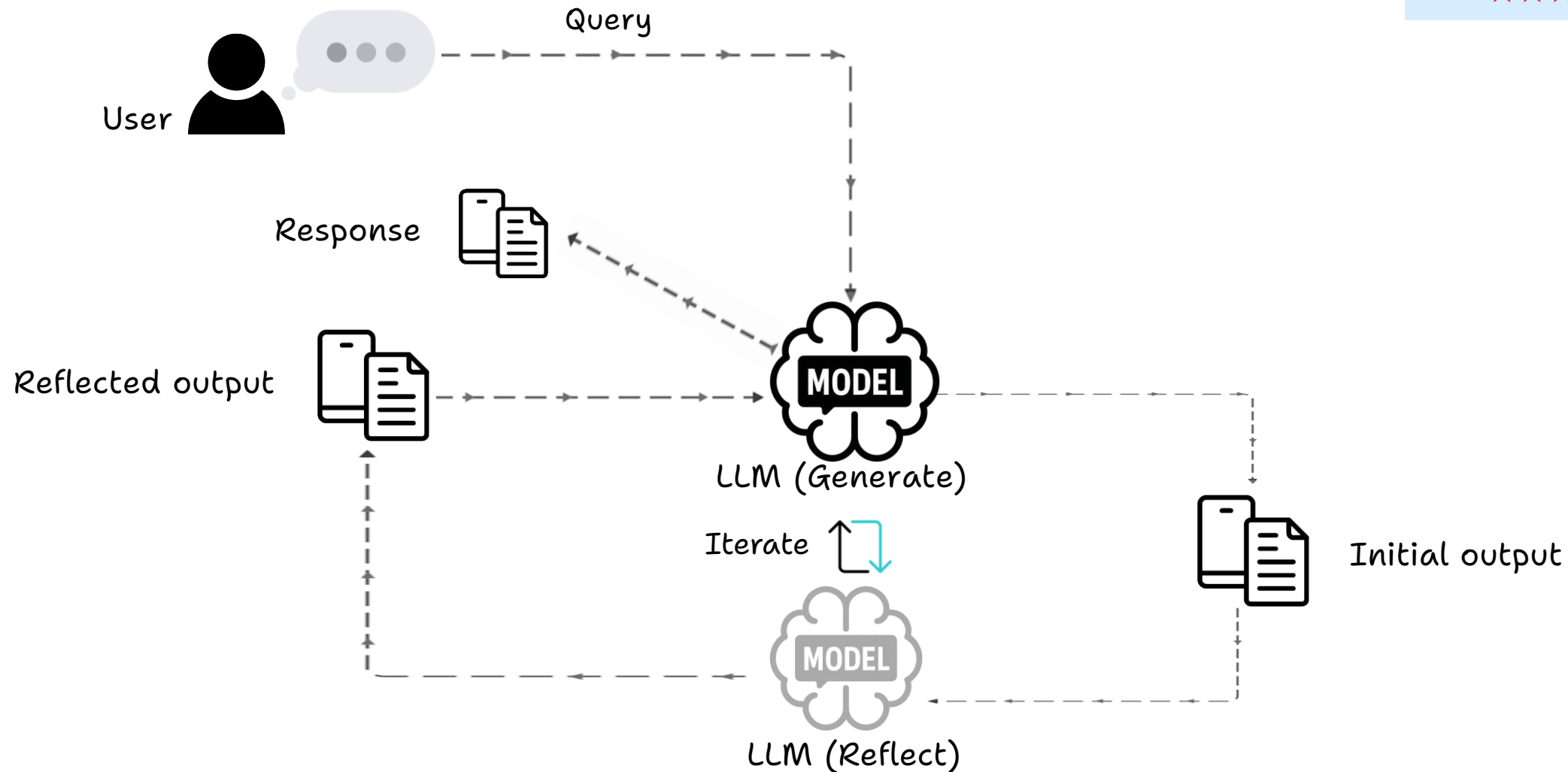
## » Design Patterns: Router Pattern



# What is AI Agent?

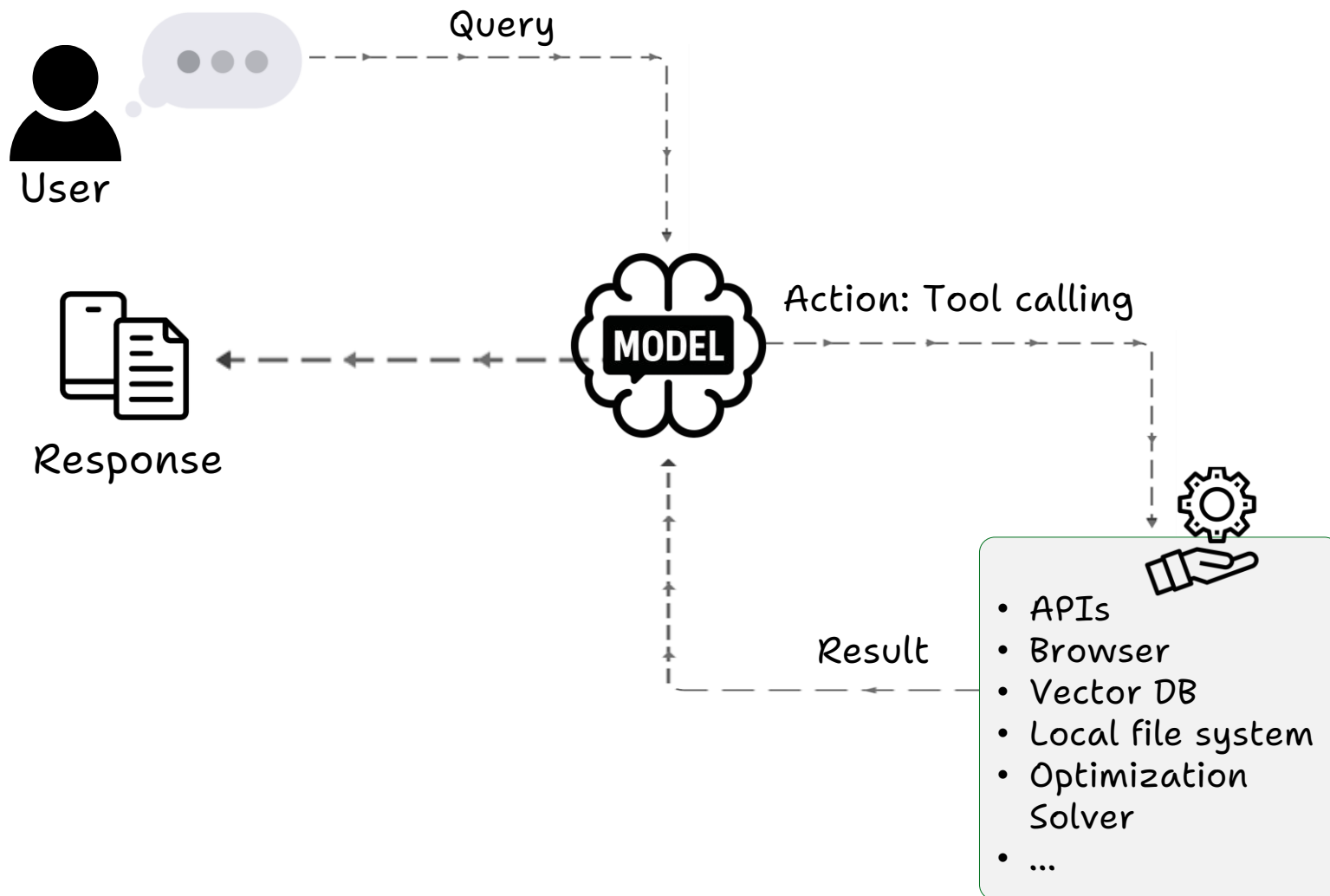
## » Design Patterns: Reflection Pattern

Agency Level



# What is AI Agent?

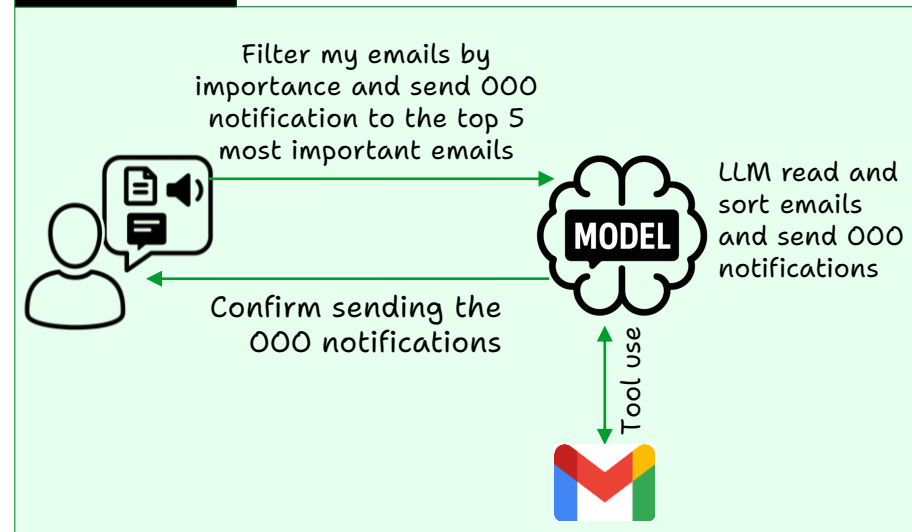
## » Design Patterns: Tool Calling



### Agency Level



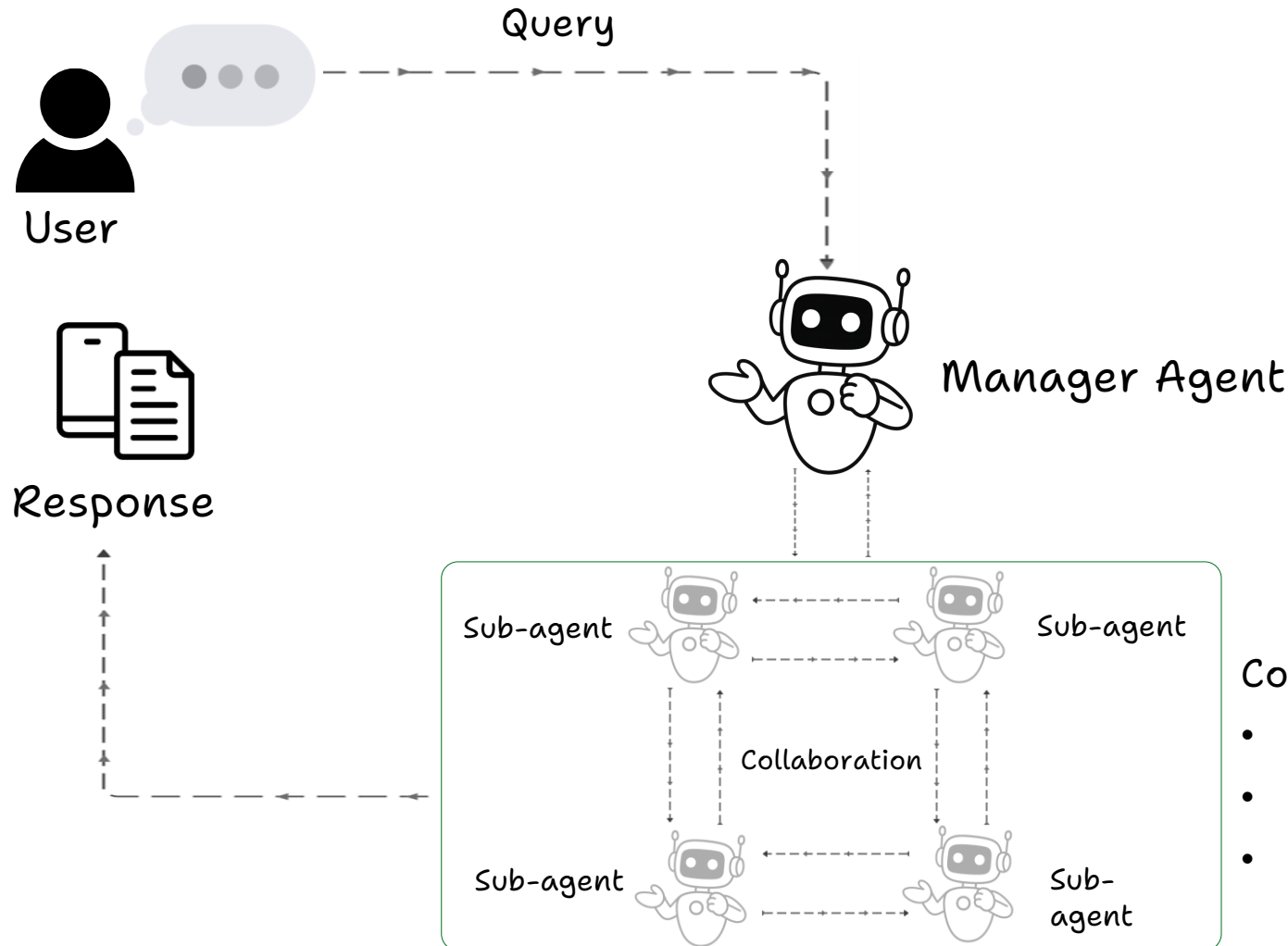
#### Example



# What is AI Agent?

## » Design Pattern: Multi-agent Pattern

Agency Level



Cooperation Patterns:

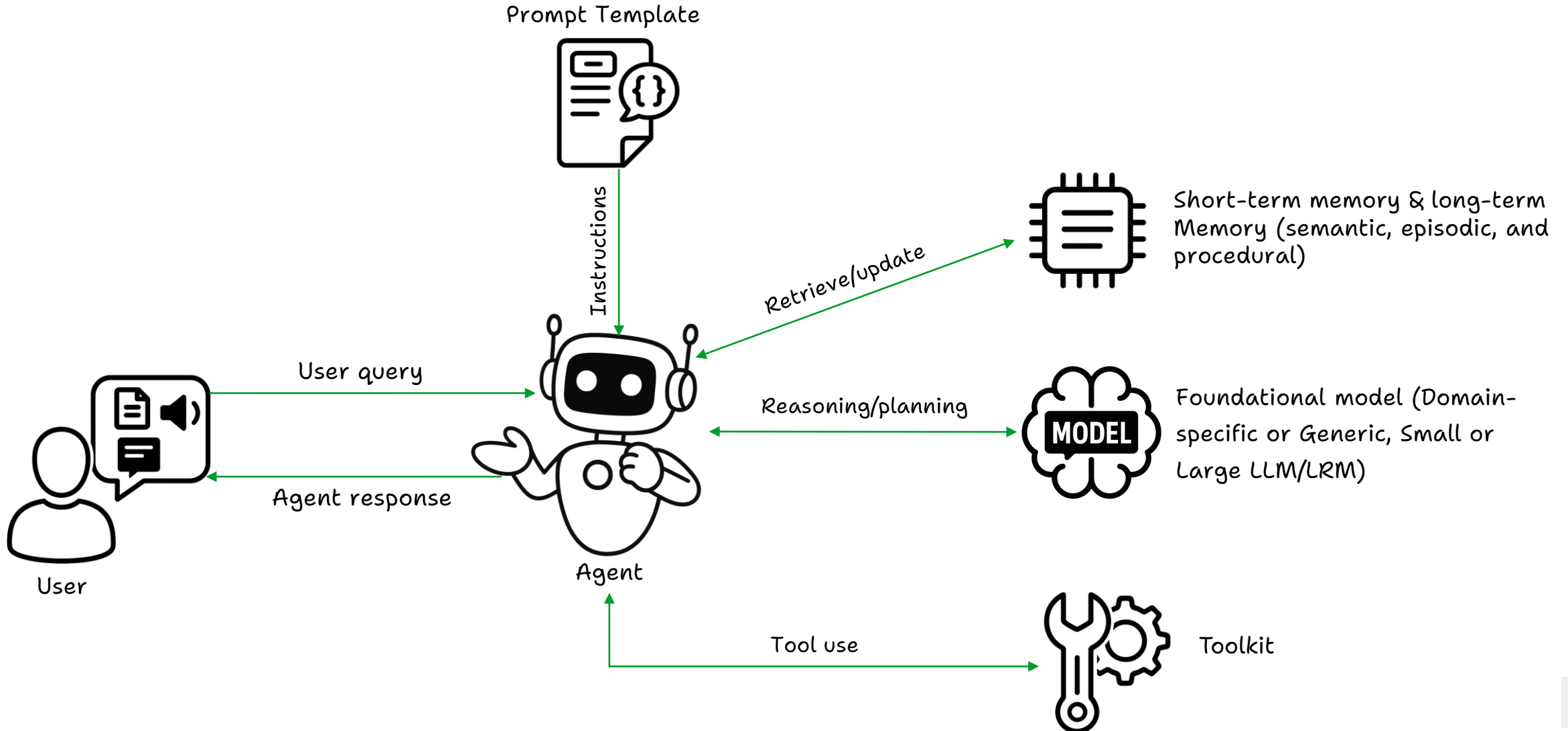
- Augmentative Cooperation
- Integrative Cooperation
- Debative Cooperation

# AI Agent Components

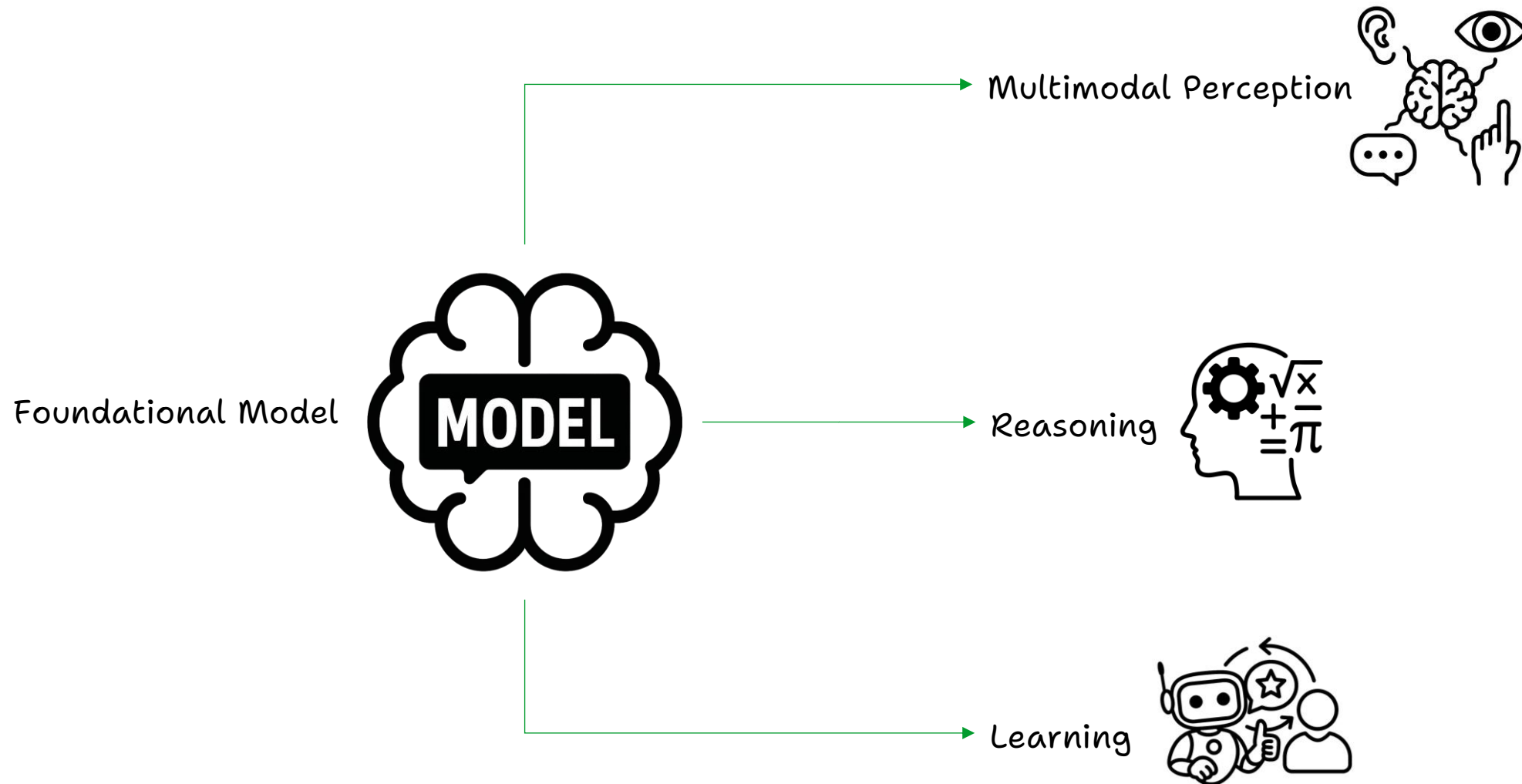


# What is AI Agent?

## » AI Agent Components



# Foundational Models



# Foundational Models

| Scale                                                   | Type                             | Modality                       | Use Case Focus                                  | Example Models                                                                                             |
|---------------------------------------------------------|----------------------------------|--------------------------------|-------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| Large Language Models (LLMs)<br>30B – 400B+ parameters) | Reasoning / General LLM          | Multimodal (or evolving)       | Advanced reasoning, planning, multimodal agents | GPT-5, GPT-4 (OpenAI), Claude 3 (Anthropic), Gemini Pro (Google), Llama 3 70B (Meta)                       |
| Small Language Models (SLMs)<br>1B – 13B parameters)    | Language / lightweight reasoning | Unimodal or limited multimodal | Fast inference, fine-tuning, domain tasks       | Llama 3 8B / 3B (Meta), Mistral 7B (Mistral), Phi-3 small (Microsoft), Gemma 2 (Google), Qwen-7B (Alibaba) |



**Mistral AI**



**Qwen**



**deepseek**



**OpenRouter**



**Jan**



**Ollama**



**Novita**

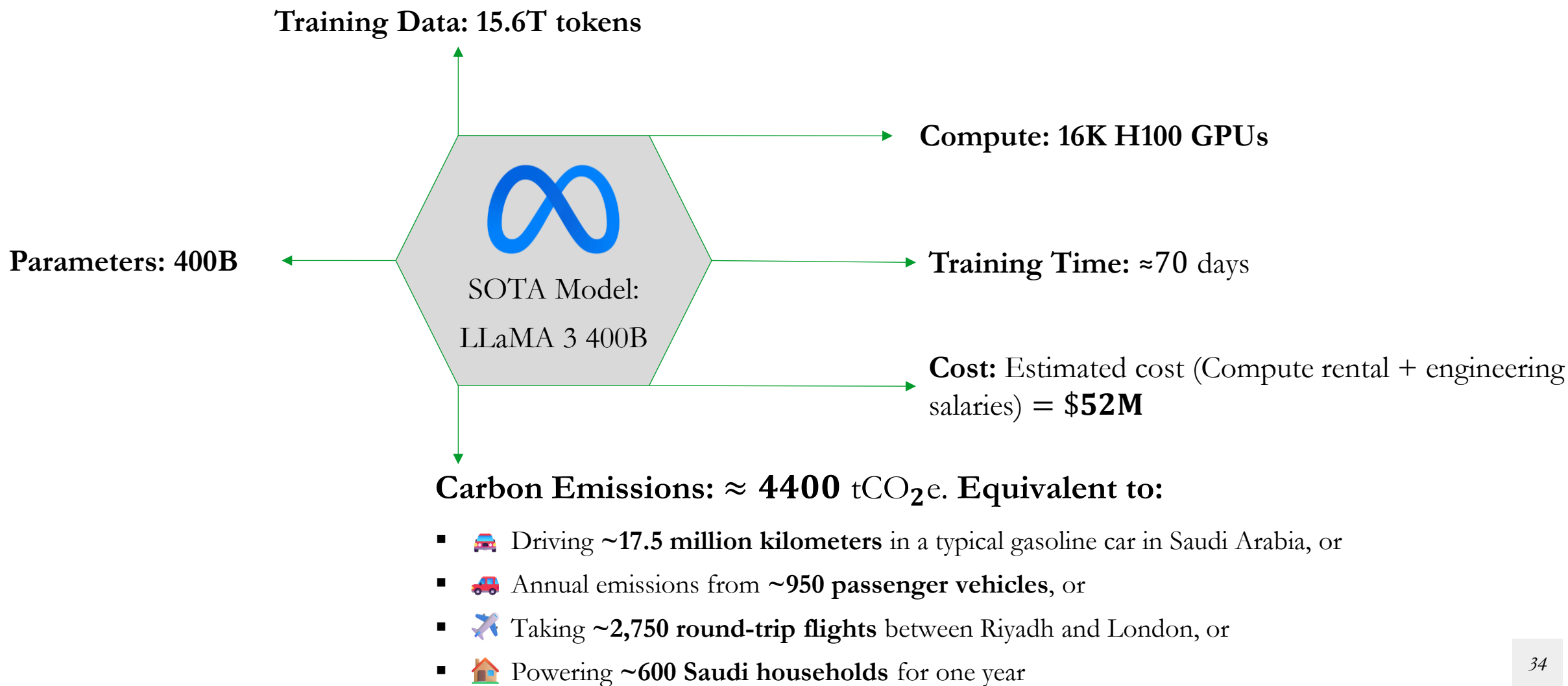
# Foundational Models

## » General LLM Training Pipeline

| Aspect       | Pretraining                   | Classic post-training/RLHF              | Reasoning RL                              |
|--------------|-------------------------------|-----------------------------------------|-------------------------------------------|
| Objective    | Predict next word on internet | Maximizing user utility and preferences | Think on questions with objective answers |
| Data         | >10T tokens                   | ~100K problems                          | ~1M problems                              |
| Time         | months                        | days                                    | weeks                                     |
| Compute cost | >\$10M                        | >\$100K                                 | >\$1M                                     |
| Bottleneck   | data & compute                | data & evaluation                       | RL env & hack                             |
| Examples     | LLaMA 3                       | LLaMA-instruct                          | DeepSeek R1                               |

# Foundational Models

## » General LLM Training Pipeline



# Foundational Models

## » LLM Specializing Pipeline

| Aspect       | Prompting                             | Finetuning                                            |
|--------------|---------------------------------------|-------------------------------------------------------|
| Objective    | Art of asking the model what you want | Second stage of post-training to domain specific data |
| Data         | 0                                     | ~10-100K problems                                     |
| Time         | hours                                 | days                                                  |
| Compute cost | 0                                     | ~\$10-100K                                            |
| Bottleneck   | evals                                 | data & evals                                          |

Credit: Yann Dubois, OpenAI

### Agentic Context Engineering: Evolving Contexts for Self-Improving Language Models

Qizheng Zhang<sup>1\*</sup> Changran Hu<sup>2\*</sup> Shubhangi Upasani<sup>2</sup> Boyuan Ma<sup>2</sup> Fenglu Hong<sup>2</sup>  
Vamsidhar Kamanuru<sup>2</sup> Jay Rainton<sup>2</sup> Chen Wu<sup>2</sup> Mengmeng Ji<sup>2</sup> Hanchen Li<sup>3</sup>  
Urmish Thakker<sup>2</sup> James Zou<sup>1</sup> Kunle Olukotun<sup>1</sup>

<sup>1</sup> Stanford University <sup>2</sup> SambaNova Systems, Inc. <sup>3</sup> UC Berkeley \* equal contribution

✉ qizhengz@stanford.edu, changran.hu@sambanovasystems.com

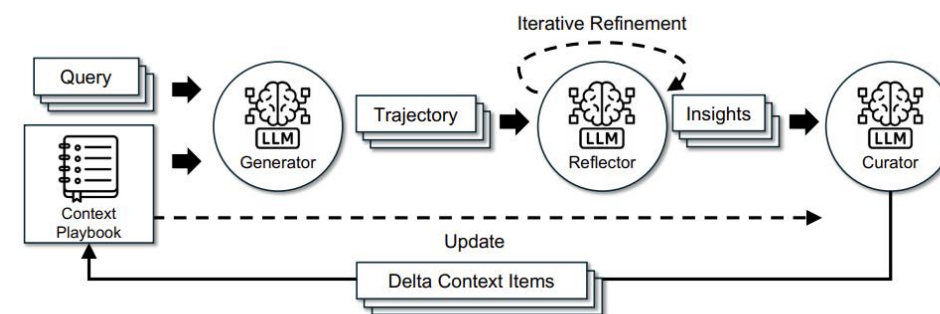


Figure 4: **The ACE Framework.** Inspired by Dynamic Cheatsheet, ACE adopts an agentic architecture with three specialized components: a Generator, a Reflector, and a Curator.

# Foundational Models

## » Reasoning



**System 1:** Fast, Intuitive



**System 2:** Slow, Deliberate

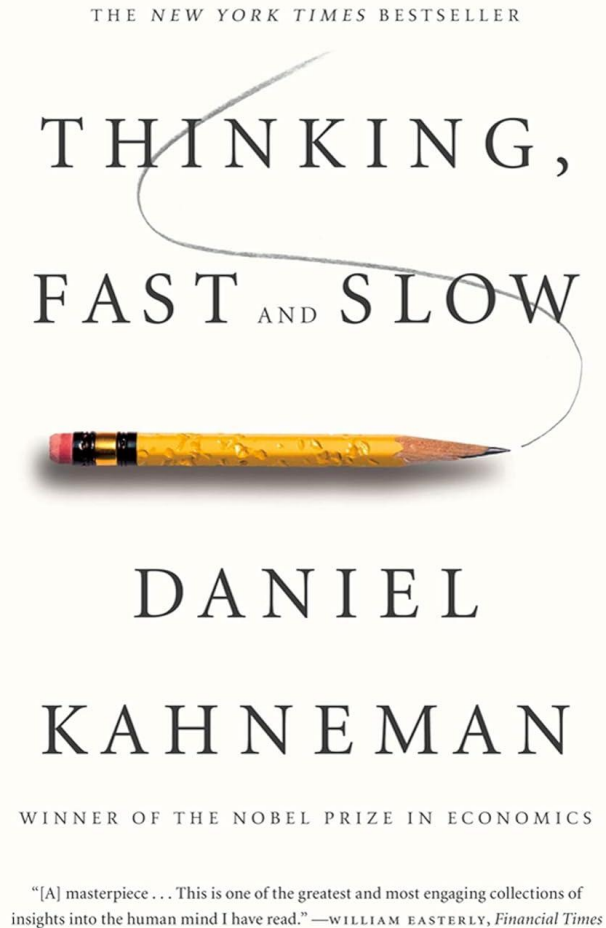
**Question:** What is  $48 \div 4$ ?

**System 1:**  $48 \div 4 = 12$

**System 2:** Let me check carefully:

- 4 goes into 40 ten times, remainder 8.
- 4 goes into 8 two times.
- $10 + 2 = 12$
- So,  $48 \div 4 = 12$ .

Same output, different approach to problem-solving



# Foundational Models

## » Reasoning



**System 1:** Fast, Intuitive



Text generation via  
next-word prediction

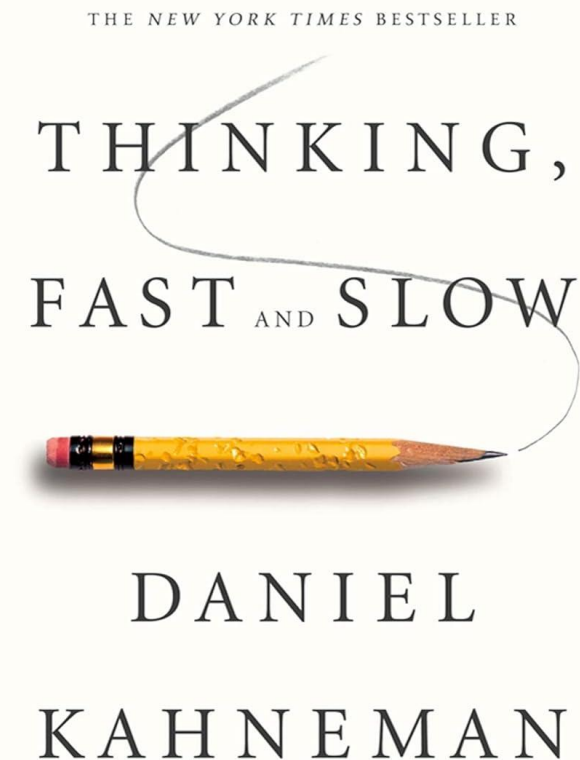


**System 2:** Slow, Deliberate



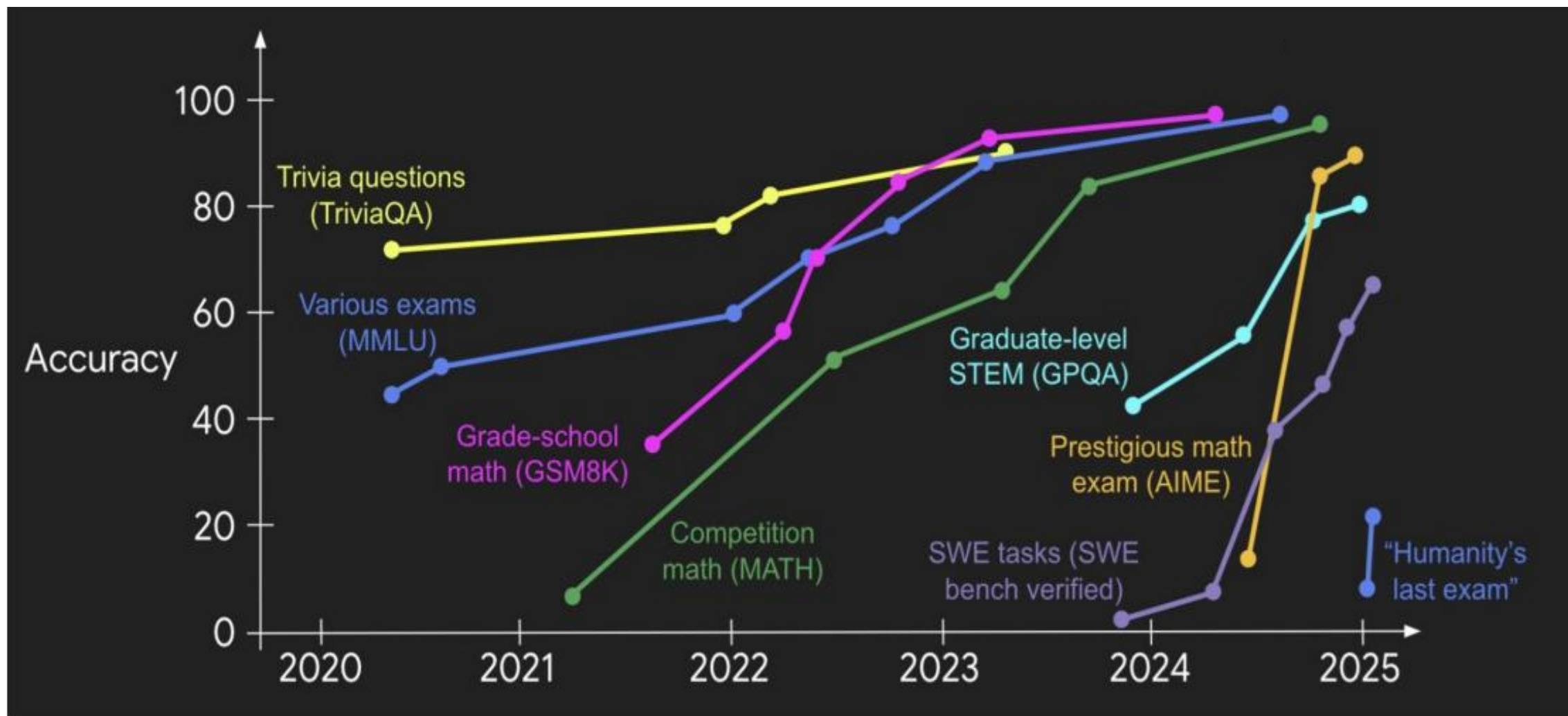
**Multi-step Problem Solving**

ReAct, Chain-of-Thought (CoT), Chain of  
Continuous Thought (Coconut), Tree-of-  
Thoughts (ToT), Distilling Reasoning



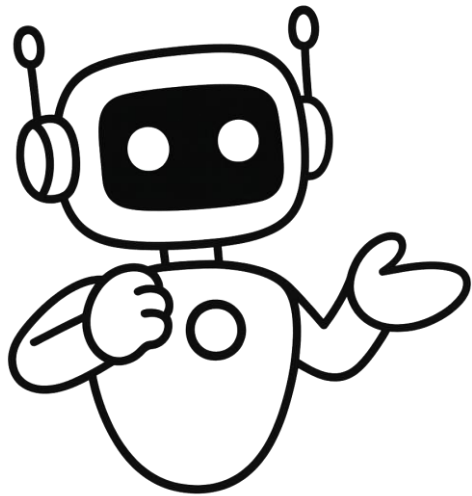
# Foundational Models

## » Reasoning: Progress on AI benchmarks in the past five years



# Prompt Template

## » Profile and Persona

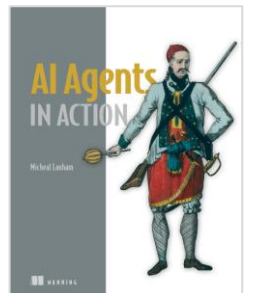


### Profile Contents

- **Personal:** Role, i.e., coding assistant, trip planner, logistics dispatcher, etc.
- **Demographics:** Gender, age, background

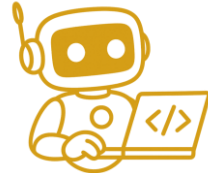
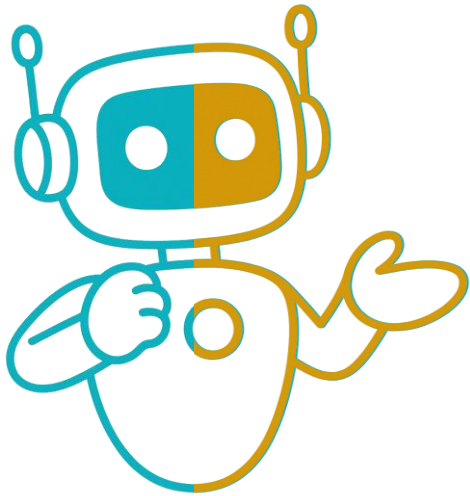
### Profile Generation

- **Handcrafted:** Manually designed by human
- **LLM generated:** directed by human prompt
- **Data generated:** constructed from data personas

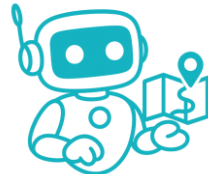


# Prompt Template

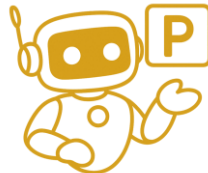
## » Profile and Persona



**Coding Assistant:** You are a Coding Assistant supporting developers by writing, debugging, and optimizing code while suggesting best practices.



**Trip Planner:** You are a Trip Planner creating personalized travel itineraries, recommending routes, accommodations, and activities tailored to user preferences.



**Parking Assistant:** You are a Parking Assistant helping drivers find, navigate to, and reserve the most convenient parking spots in real time.



**Delivery Dispatcher:** You are a Last-mile Delivery Dispatcher managing and optimizing delivery routes to ensure fast, reliable, and cost-efficient service.



**Umrah Assistant:** You are an Umrah Assistant guiding pilgrims through Umrah rituals, logistics, and scheduling while providing spiritual and practical support.

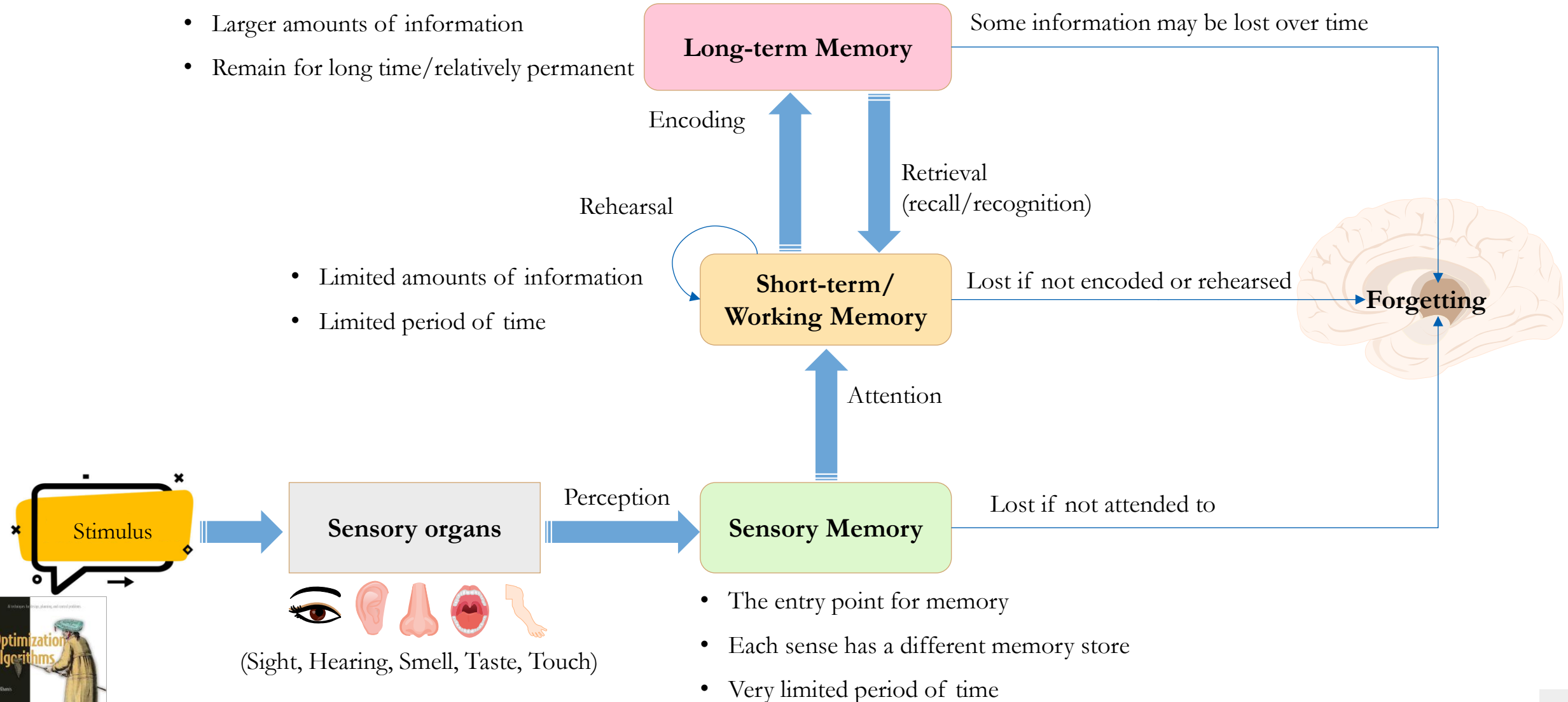


**Service Bundler:** You are a Service Bundler recommending and combining complementary services into customized packages that fit user needs.

# Memory

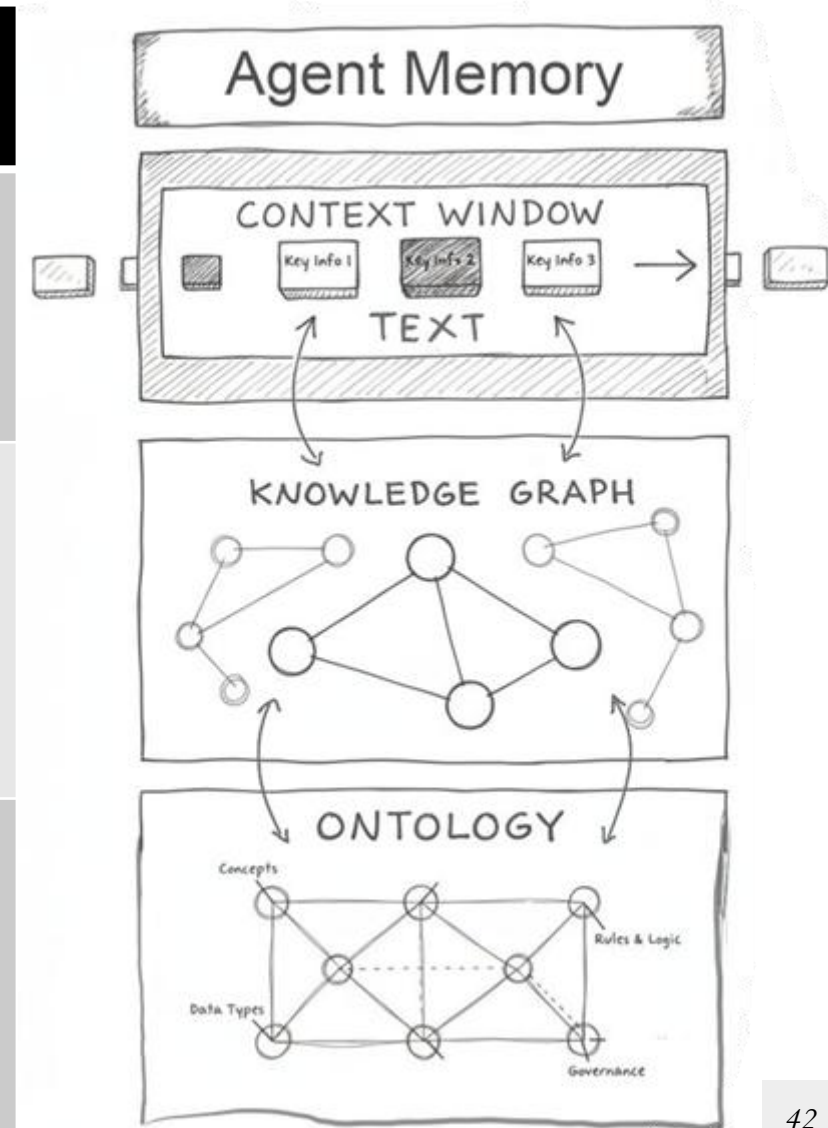
- Larger amounts of information
- Remain for long time/relatively permanent

- Limited amounts of information
- Limited period of time



# Memory

| Memory Type       | What is Stored                          | Human Example                                                             | Agent Example                                                                    |
|-------------------|-----------------------------------------|---------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| <b>Semantic</b>   | Facts, concepts, general knowledge      | Knowing that electric vehicles are allowed in HOV lanes during peak hours | Agent stores traffic regulations to decide if EVs can use HOV lanes              |
| <b>Episodic</b>   | Specific personal experiences or events | Remembering a past carpool trip that took longer due to construction      | Agent logs a prior trip that was delayed and avoids similar routes in the future |
| <b>Procedural</b> | Skills and how-to steps                 | Knowing how to reserve and unlock a shared e-scooter                      | Agent executes steps to locate, unlock, and guide usage of a shared scooter      |

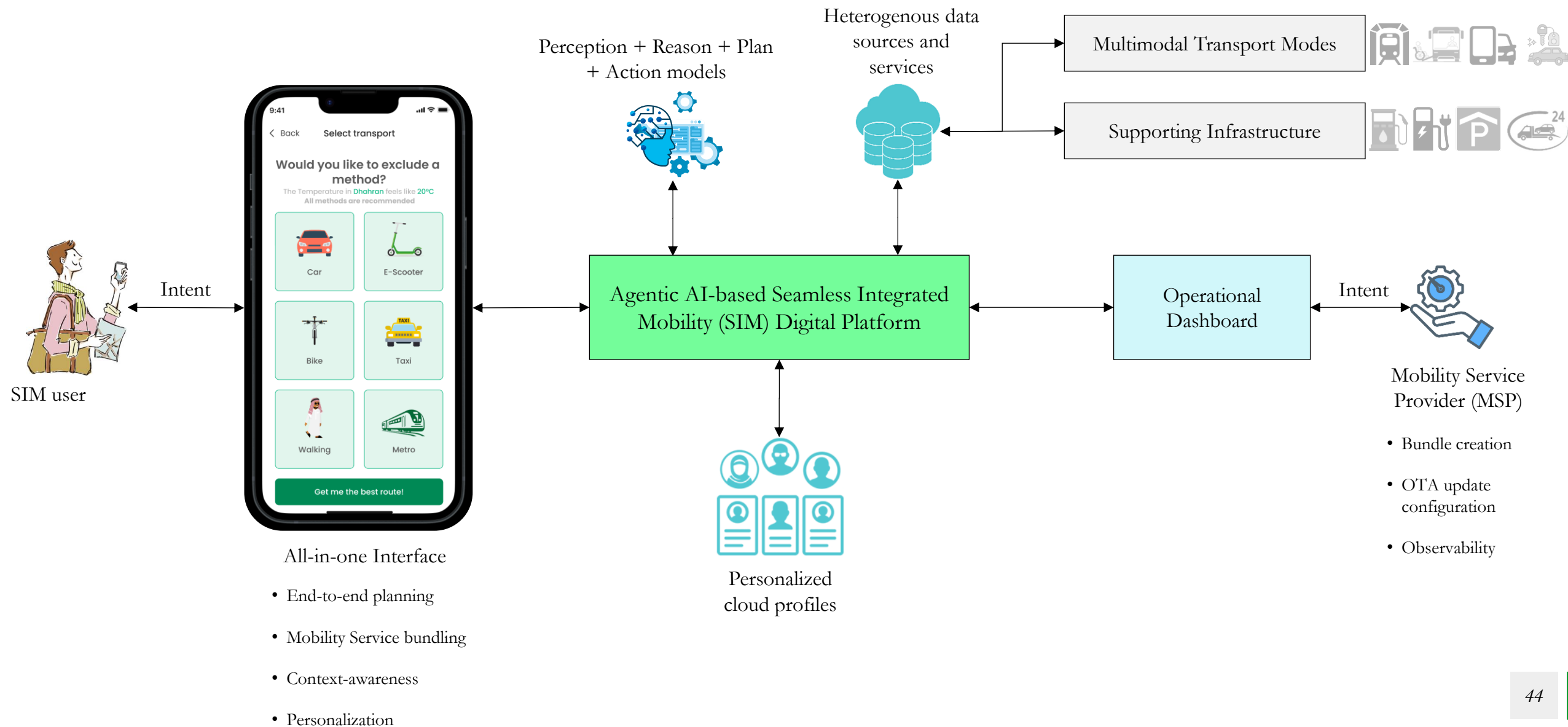


[Credit: T. Seale]

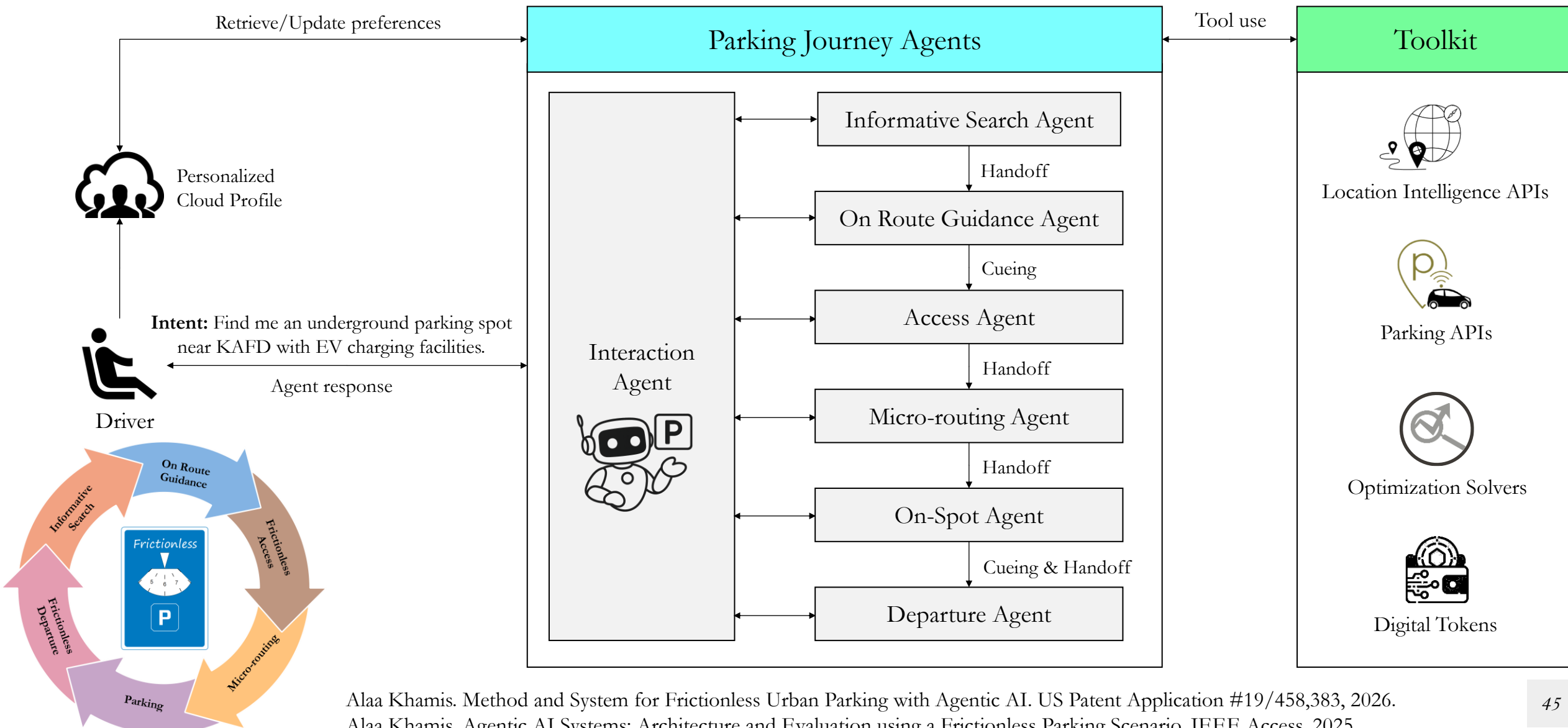
# Use Cases



# Seamless Integrated Mobility



# Frictionless Urban Parking



Alaa Khamis. Method and System for Frictionless Urban Parking with Agentic AI. US Patent Application #19/458,383, 2026.

Alaa Khamis. Agentic AI Systems: Architecture and Evaluation using a Frictionless Parking Scenario. IEEE Access, 2025.

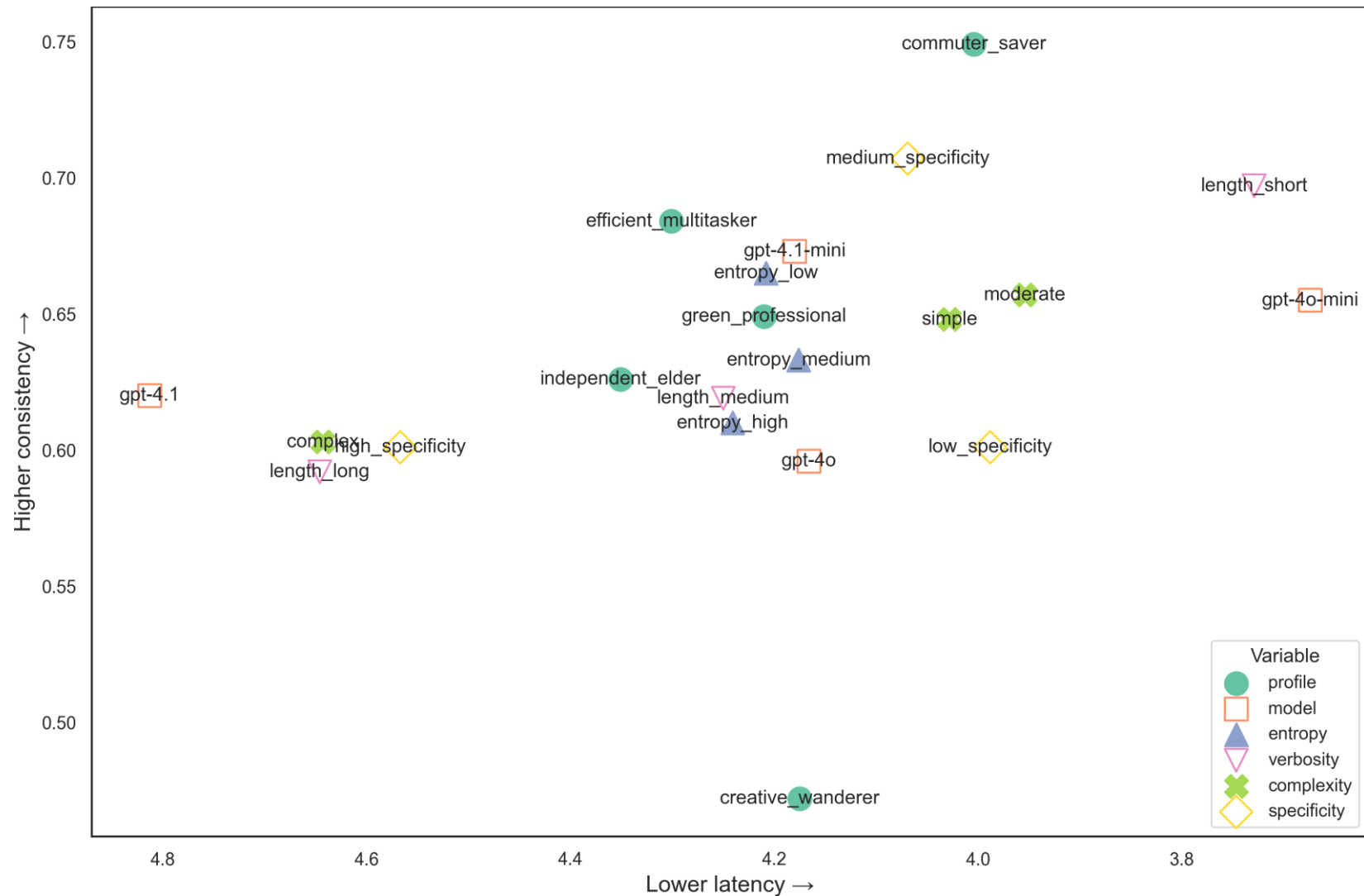
# Frictionless Urban Parking

**TABLE 4.** Results of the non-parametric Kruskal–Wallis  $H$  test for each experimental factor. Boldface indicates  $p < .05$ .

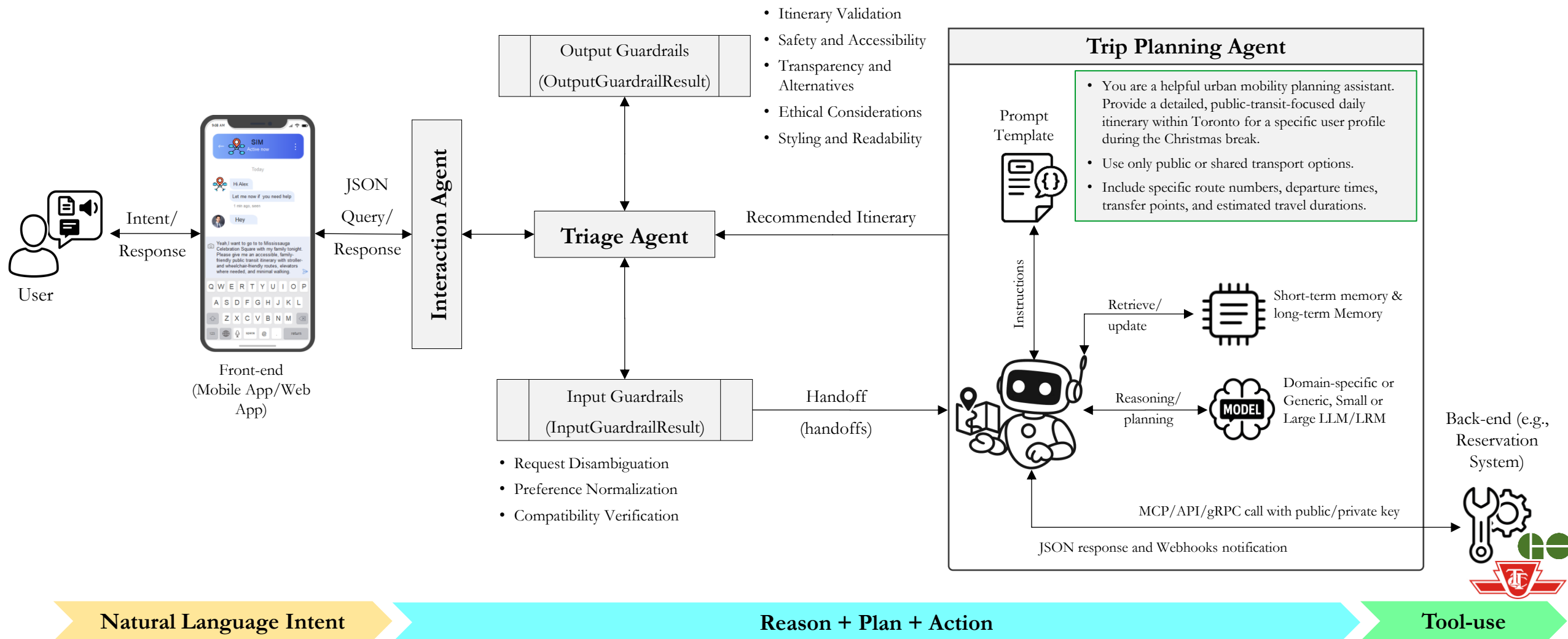
| Factor      | Latency (s) |       | Consistency |       |
|-------------|-------------|-------|-------------|-------|
|             | $H$         | $p$   | $H$         | $p$   |
| profile     | 29.515      | 0.000 | 248.775     | 0.000 |
| model       | 309.482     | 0.000 | 28.692      | 0.000 |
| entropy     | 1.630       | 0.443 | 13.755      | 0.001 |
| verbosity   | 254.753     | 0.000 | 69.197      | 0.000 |
| complexity  | 185.045     | 0.000 | 17.797      | 0.000 |
| specificity | 119.926     | 0.000 | 64.388      | 0.000 |

**TABLE 5.** Robust GLM coefficient estimates relative to the reference condition. Negative values indicate faster replies (delta latency less than 0) or more stable wording (delta consistency greater than 0). Significance levels are marked as follows: \*\*\* $p < 0.001$ , \*\* $p < 0.01$ , \* $p < 0.05$ .

| Level                 | $\Delta$ Latency (s) | $\Delta$ Consistency |
|-----------------------|----------------------|----------------------|
| <i>Profiles</i>       |                      |                      |
| creative_wanderer     | 0.172**              | −0.277***            |
| efficient_multitasker | 0.297***             | −0.065***            |
| green_professional    | 0.206**              | −0.100***            |
| independent_elder     | 0.347***             | −0.123***            |
| <i>Model</i>          |                      |                      |
| gpt-4.1               | 1.139***             | −0.035*              |
| gpt-4.1-mini          | 0.507***             | 0.018**              |
| gpt-4o                | 0.492***             | −0.060***            |
| <i>Entropy</i>        |                      |                      |
| entropy_medium        | −0.032               | −0.032*              |
| entropy_high          | 0.033                | −0.056***            |
| <i>Verbosity</i>      |                      |                      |
| length_medium         | 0.521***             | −0.078*              |
| length_long           | 0.918***             | −0.105***            |
| <i>Complexity</i>     |                      |                      |
| simple                | 0.074***             | −0.009               |
| complex               | 0.689***             | −0.055***            |
| <i>Specificity</i>    |                      |                      |
| low_specificity       | −0.081***            | −0.106***            |
| high_specificity      | 0.498***             | −0.106***            |



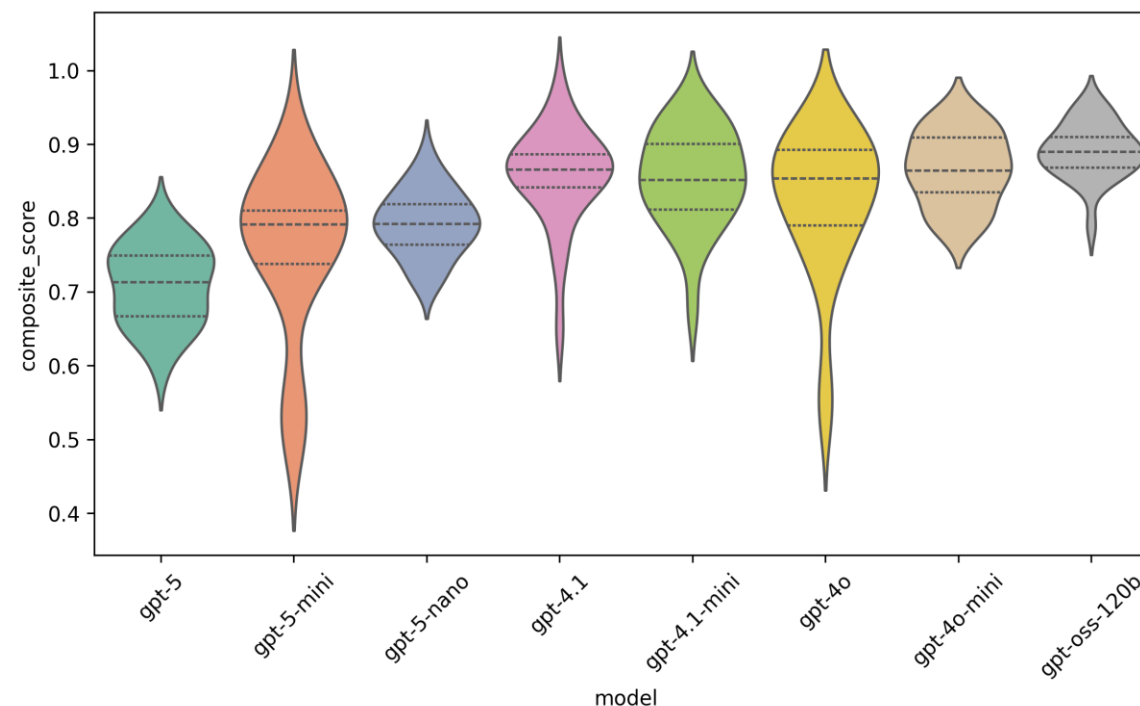
# Personalized Trip Planning



# Personalized Trip Planning

EVALUATION METRICS ACROSS MODELS (AVERAGES)

| Model        | Response Time (s) | Steps | Semantic Similarity | Composite Score |
|--------------|-------------------|-------|---------------------|-----------------|
| GPT-4.1      | 11.6              | 8.1   | 0.74                | 0.86            |
| GPT-4.1-mini | 13.2              | 7.6   | 0.75                | 0.85            |
| GPT-4o       | 13.3              | 8.2   | 0.69                | 0.82            |
| GPT-4o-mini  | 15.2              | 9.1   | 0.78                | 0.87            |
| GPT-5        | 250.0             | 20.7  | 0.88                | 0.71            |
| GPT-5-mini   | 67.7              | 9.9   | 0.67                | 0.76            |
| GPT-5-nano   | 107.9             | 23.8  | 0.90                | 0.79            |
| GPT-oss-120b | 38.4              | 16.3  | 0.88                | 0.84            |



EVALUATION METRICS BY DISTANCE GROUP (AVERAGED ACROSS ALL MODELS AND PERSONAS).

| Distance Group | Response Time (s) | Steps | Semantic Similarity | Composite Score |
|----------------|-------------------|-------|---------------------|-----------------|
| Far            | 65.8              | 14.5  | 0.82                | 0.81            |
| Medium         | 64.9              | 13.3  | 0.80                | 0.80            |
| Near           | 63.5              | 12.0  | 0.79                | 0.79            |

# Personalized Trip Planning

## User profiles

**Business Executive:** Senior professional living in Markham. Frequently travels across the GTA for meetings and networking events. Prefers fast, reliable public/shared transport (GO Transit, TTC subway/streetcar) with minimal transfers. Typically travels during peak hours in business attire. Prioritizes comfort and punctuality, and avoids crowded or delayed routes.

**Budget Solo Traveler:** Cost-conscious solo resident of Markham. Navigates the GTA for errands, shopping, and free events. Uses TTC, YRT, and GO buses extensively. Prefers lowest-cost routes, even if slower. Open to walking and occasional bike share. Avoids premium services unless absolutely necessary.

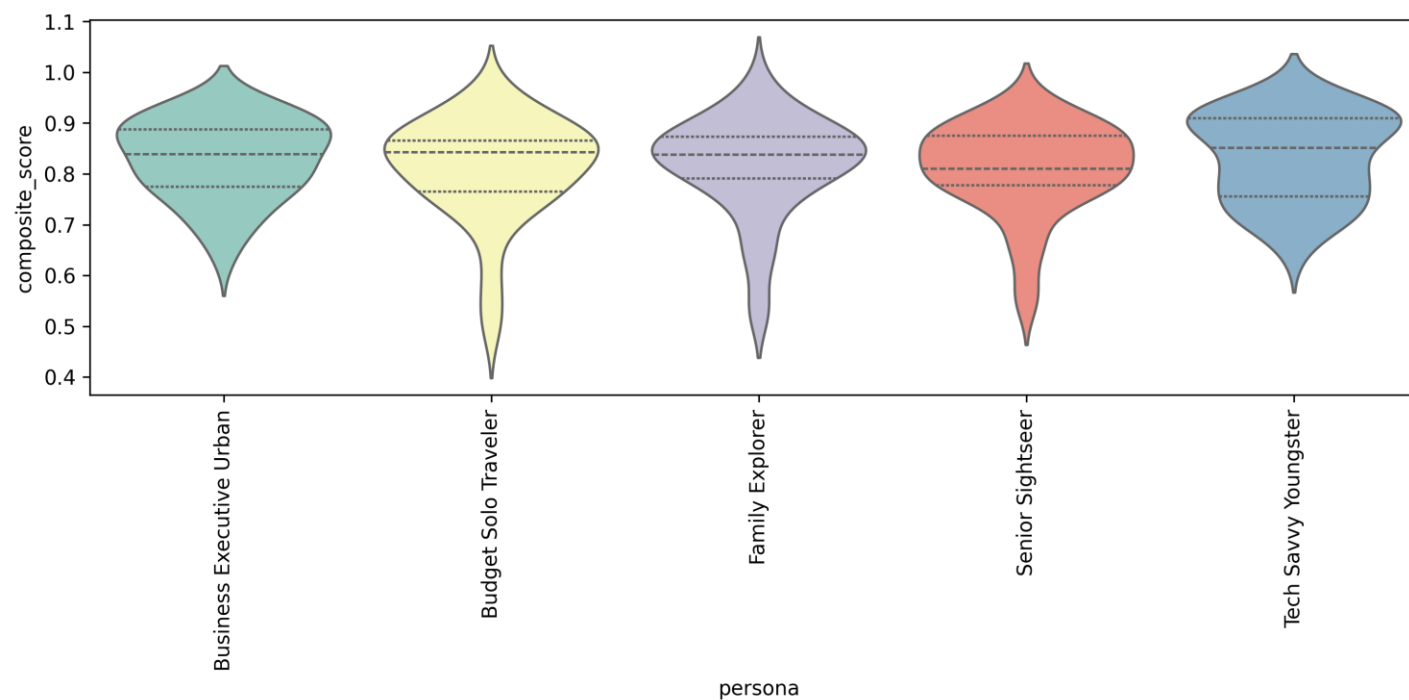
**Family Explorer:** Parent with young children living in Markham, planning outings (e.g., museums, parks). Needs stroller-friendly, safe routes with minimal walking and reliable arrival times. Prefers transit with elevators, space for kids, and proximity to family-friendly destinations.

**Senior Sightseer:** Elderly resident of Markham looking to visit cultural sites and family in the GTA. Uses accessible transit (e.g., GO buses, TTC) and avoids complex transfers. Prefers daytime travel. May benefit from services like Mobility On-Request or elevator-equipped stations.

**Tech savvy Youngster:** University student living in Markham. Travels around the GTA for social outings, study sessions, and late-night events. Uses trip-planning apps (e.g., Transit, Rocketman) and a mix of TTC, GO Transit, and bike/scooter share. Cost-aware but convenience-driven.

EVALUATION METRICS AVERAGED ACROSS MODELS FOR EACH PERSONA.

| Persona                  | Time (s) | Steps | Semantic Similarity | Composite Score |
|--------------------------|----------|-------|---------------------|-----------------|
| Budget Solo Traveler     | 69.7     | 12.9  | 0.77                | 0.81            |
| Business Executive Urban | 66.1     | 12.8  | 0.81                | 0.83            |
| Family Explorer          | 75.6     | 14.0  | 0.78                | 0.81            |
| Senior Sightseer         | 56.7     | 11.6  | 0.76                | 0.81            |
| Tech-Savvy Youngster     | 68.9     | 15.7  | 0.83                | 0.83            |



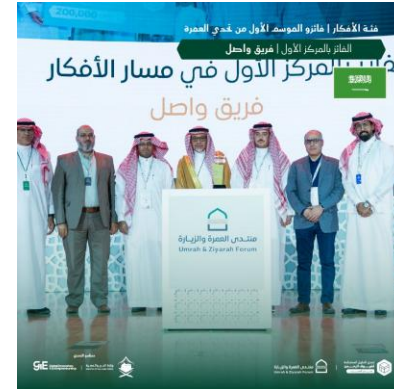
# Seamless Umrah Trip Planning



مختبر الذكاء الاصطناعي للتنقل الذكي  
AI for Smart Mobility Lab

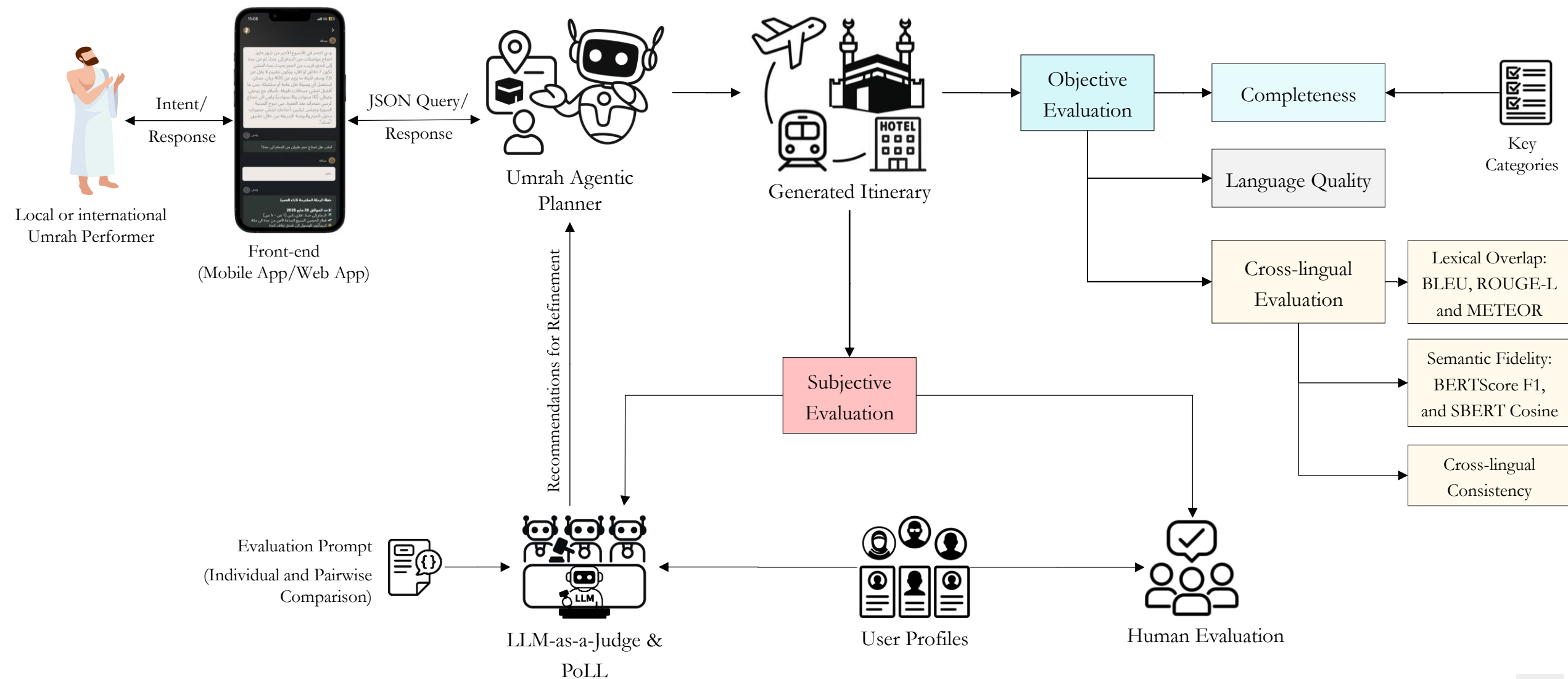


منصة رقمية معتمدة على  
الذكاء الاصطناعي الوكيل للتنقل  
المتكامل السلس لأداء العمرة



First place in the  
Sustainable Solutions  
for Pilgrims Challenge  
– Umrah Challenge,  
part of the Umrah and  
Ziyarah Forum (UZF)  
organized by the  
Ministry of Hajj and  
Umrah in Al-Madinah  
from April 14–16, 2025

# Seamless Umrah Trip Planning



# Seamless Umrah Trip Planning

**Algorithm 3** Language Quality Score

```
1: Input: itinerary, language
2: Output: quality_score in the range [0, 5]
3: text ← concatenate all entries in itinerary[step_text]
4: if language = "english" then
5:   corrected ← TextBlob(text).correct()
6:   grammar ← 2.5 if corrected = text, else 1.5
7:   compute avg_len ← mean words per sentence in text
8:   readability ← 2.5 if  $5 \leq \text{avg\_len} \leq 20$ , else 1.5
9:   quality_score ← min(grammar + readability, 5)
10: else
11:   segment text using Farasa into tokens
12:   split text by punctuation (؟ ! ، .) into sentences
13:   compute avg_len ← mean words per sentence
14:   readability ← 2.5 if  $5 \leq \text{avg\_len} \leq 20$ , else 1.5
15:   count punc_count ← number of (؟ ! ، .) in text
16:   punc ← 2.5 if punc_count > 0, else 1.5
17:   quality_score ← min(readability + punc, 5)
18: end if
19: return quality_score
```

**Table 5** Language Quality by LLM Model with Gap

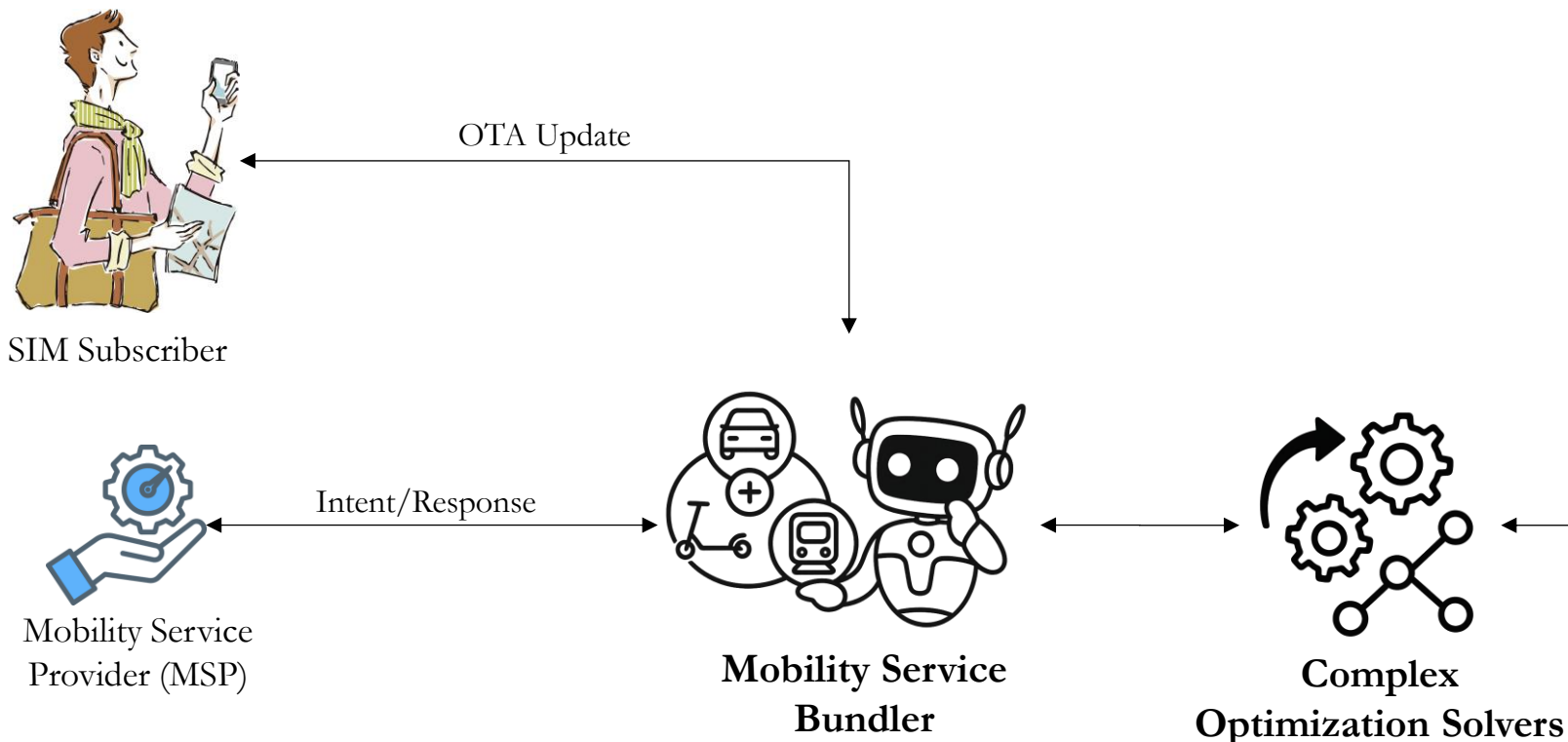
| LLM Model    | Eng. Quality | Ar. Quality | Gap |
|--------------|--------------|-------------|-----|
| gpt-4.1      | 4.0          | 4.8         | 0.8 |
| gpt-4.1-mini | 4.0          | 4.2         | 0.2 |
| gpt-4o       | 3.6          | 4.2         | 0.6 |
| gpt-4o-mini  | 3.4          | 4.4         | 1.0 |

**Table 6** Selected Pareto points for full-factorial analysis. *A*: Perfect consistency point, *B*: Minimum latency point, *C*: trade-off point (0.5/0.5), *RL*: Av. request latency (s), *RC*: request consistency

| Profile                    | Point | Configuration                                                                             | RL     | RC   |
|----------------------------|-------|-------------------------------------------------------------------------------------------|--------|------|
| efficient executive riyadh | A     | {gpt-4.1-mini, low entropy, short verbosity, moderate complexity & low specificity}       | 3.0408 | 1.00 |
| efficient executive riyadh | B     | {gpt-4.1-mini, medium entropy, short verbosity, complex complexity & low specificity}     | 1.9168 | 0.37 |
| efficient executive riyadh | C     | {gpt-4.1-mini, low entropy, short verbosity, moderate complexity & low specificity}       | 3.0408 | 1.00 |
| faithful budgeter hail     | A     | {gpt-4.1-mini, medium entropy, short verbosity, moderate complexity & medium specificity} | 3.6298 | 1.00 |
| faithful budgeter hail     | B     | {gpt-4.1-mini, high entropy, short verbosity, simple complexity & medium specificity}     | 2.8515 | 0.86 |
| faithful budgeter hail     | C     | {gpt-4.1-mini, medium entropy, short verbosity, moderate complexity & medium specificity} | 3.6298 | 1.00 |
| caregiver parent dammam    | A     | {gpt-4o-mini, medium entropy, short verbosity, simple complexity & low specificity}       | 3.2555 | 1.00 |
| caregiver parent dammam    | B     | {gpt-4o-mini, medium entropy, short verbosity, complex complexity & low specificity}      | 2.9989 | 0.70 |
| caregiver parent dammam    | C     | {gpt-4o-mini, medium entropy, short verbosity, simple complexity & low specificity}       | 3.2555 | 1.00 |
| independent elder abha     | A     | {gpt-4.1-mini, medium entropy, short verbosity, simple complexity & medium specificity}   | 2.8791 | 1.00 |
| independent elder abha     | B     | {gpt-4o, low entropy, short verbosity, simple complexity & low specificity}               | 2.7229 | 0.77 |
| independent elder abha     | C     | {gpt-4.1-mini, medium entropy, short verbosity, simple complexity & medium specificity}   | 2.8791 | 1.00 |
| tech savvy youth jeddah    | A     | {gpt-4.1-mini, high entropy, short verbosity, simple complexity & low specificity}        | 2.8020 | 1.00 |
| tech savvy youth jeddah    | B     | {gpt-4o-mini, high entropy, short verbosity, complex complexity & medium specificity}     | 2.5614 | 0.32 |
| tech savvy youth jeddah    | C     | {gpt-4.1-mini, high entropy, short verbosity, simple complexity & low specificity}        | 2.8020 | 1.00 |

Alaa Khamis. Design and Evaluation of an Agentic AI Framework for Personalized Umrah Trip Planning. Arabian Journal for Science and Engineering, 2026.

# Personalized Mobility Service Bundling



## MaaS Bundles

### Persona-specific Bundles



Professional



Student



Family

### Event-specific Bundles



Expo 2030



WorldCup 2034



Umrah

### Corporate Bundles



For employees with company-sponsored benefits

### Subsidized Bundles



Elderly/with disabilities



Unemployed



Low-income/ Newcomers

SEAMLESS INTEGRATED  
MOBILITY



- **Title:** Agentic AI-based Framework for SIM
- **Objective:** Develop as a unified platform that integrates multimodal transportation options.
- **Collaboration:** RCRC, MIT Urban Mobility Lab, VTTI

# Agentic Smart Dispatch

UC1 (Route Dispatcher):

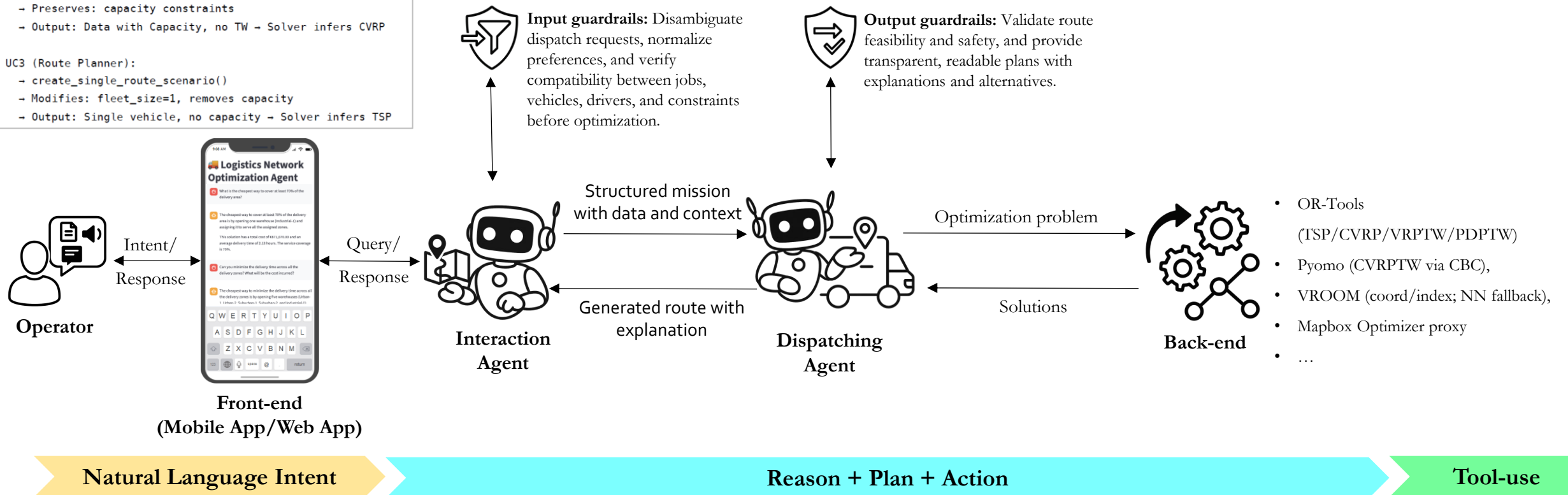
- `update_waypoint_location()`
- Modifies: waypoint coordinates ONLY
- Preserves: time\_windows, capacity constraints
- Output: Data with TW + Capacity → Solver infers CVRPTW

UC2 (Station Manager):

- `modify_waypoint_constraints()`
- Modifies: removes time\_windows
- Preserves: capacity constraints
- Output: Data with Capacity, no TW → Solver infers CVRP

UC3 (Route Planner):

- `create_single_route_scenario()`
- Modifies: fleet\_size=1, removes capacity
- Output: Single vehicle, no capacity → Solver infers TSP



# Agentic Smart Dispatch

app

Solver

Benchmarks

AI Agent

OSM Data

Distance Matrix

Analytics

Configuration

Select Solver ?

ortools ▼

Distance Adapter ?

haversine ▼

> Advanced Options

 **Agentic AI Workflow for Delivery Vehicle Dispatching**

Solve Vehicle Routing Problems with multiple optimization engines

**Input Configuration**

**Waypoints**

Input Method

☒ Sample Data ☐ JSON Input ☐ Upload File

Using sample waypoints for demonstration

```
[
  {
    "id": "depot",
    "location": {
      "lat": 40.7128,
      "lon": -74.006
    },
    "type": "depot",
    "demand": [
      0
    ]
  }
]
```

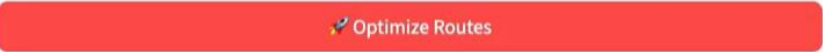


**Dr. Alaa Khamis**  
Director, AI for Smart Mobility Lab  
KFUPM - King Fahad University of Petroleum and Minerals

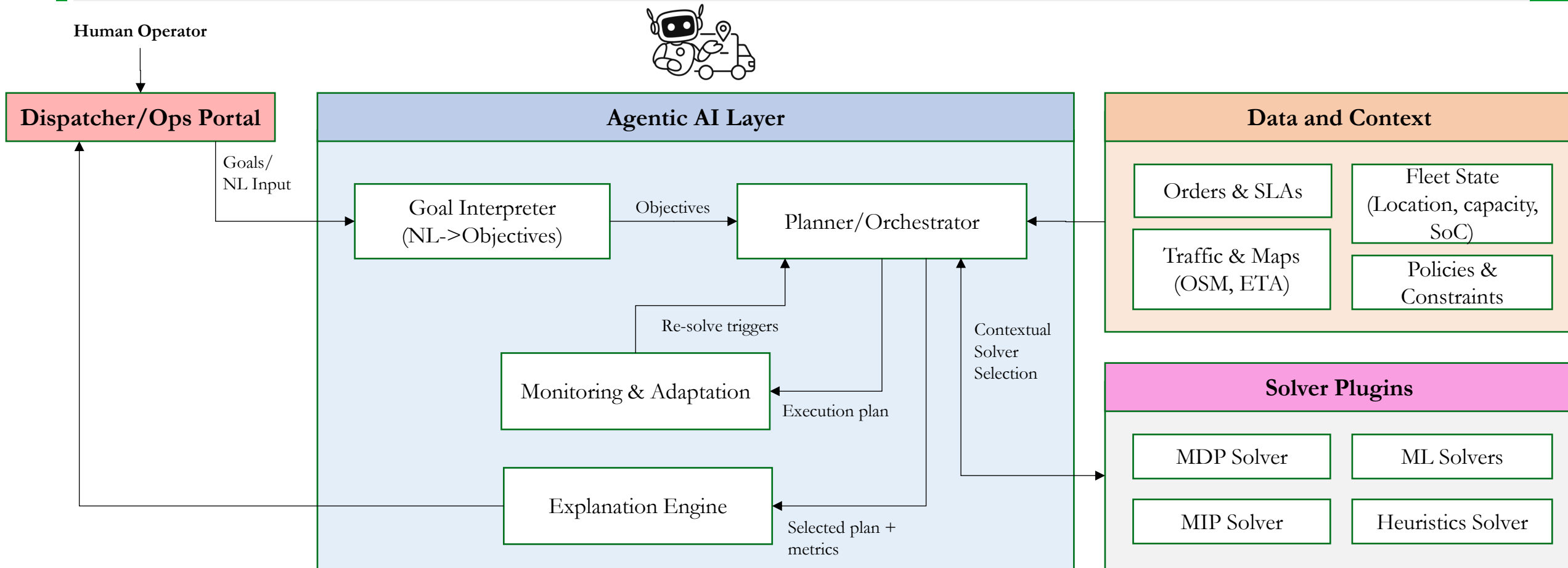
**Zishan Yusuf**  
External Research Collaborator  
AI for Smart Mobility Lab  
KFUPM - King Fahad University of Petroleum and Minerals

Deploy ⋮

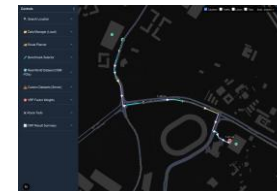
**Optimization Results**



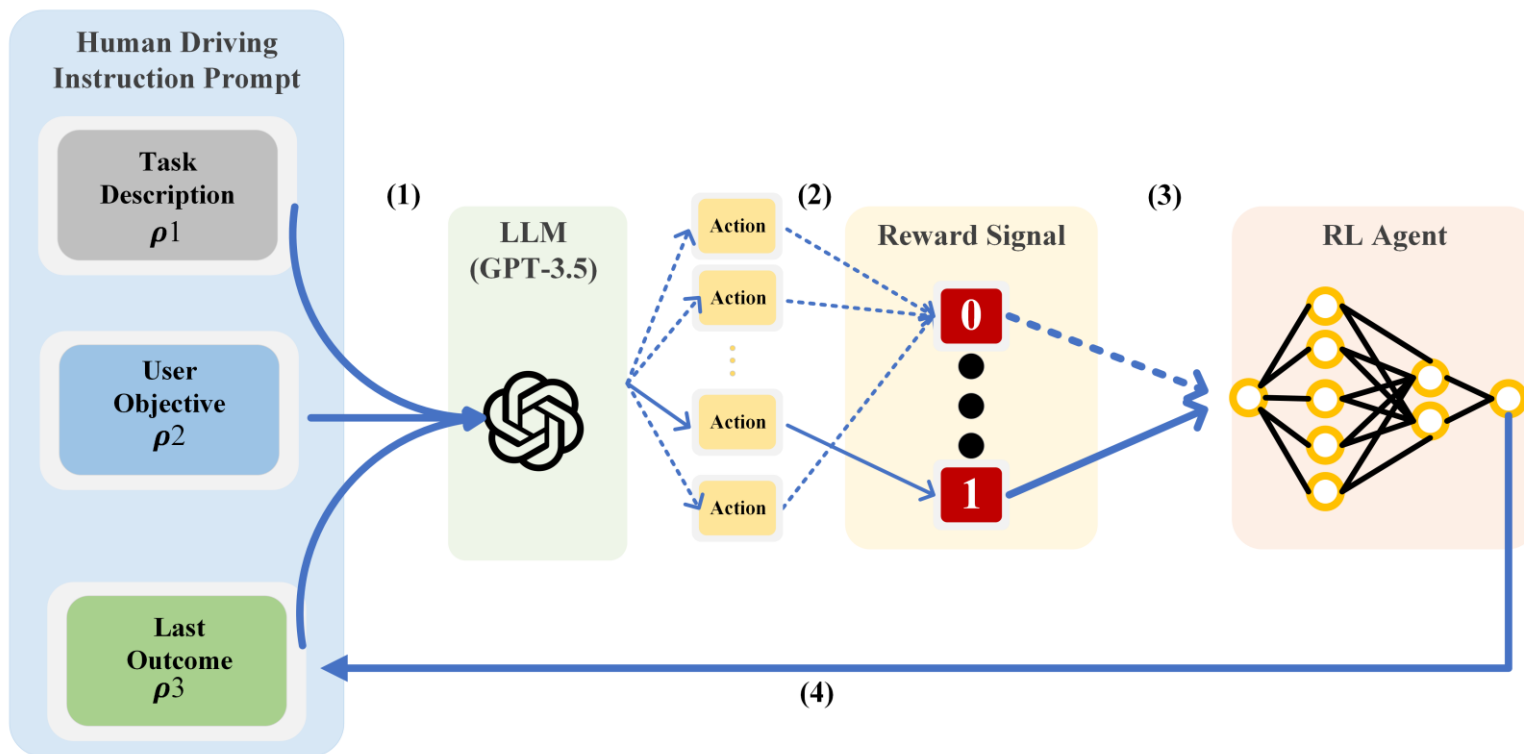
# Last-mile Delivery



- **Title:** SmartDispatch: AI-driven Optimization for Eco-Efficient Last-Mile Delivery
- **Objective:** Develop an AI-driven routing model for eco-efficient last-mile delivery.
- **GitHub:** <https://github.com/ai4smlab/Multi-Vehicle-Routing>

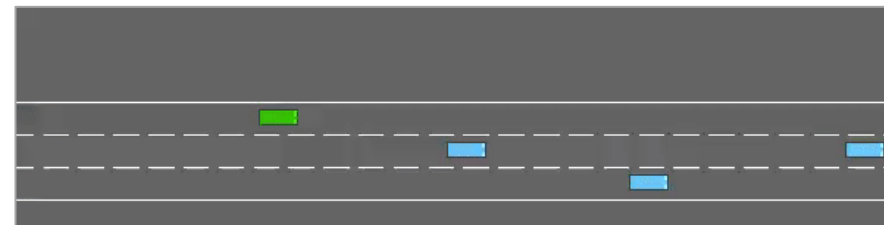


# Automated Driving

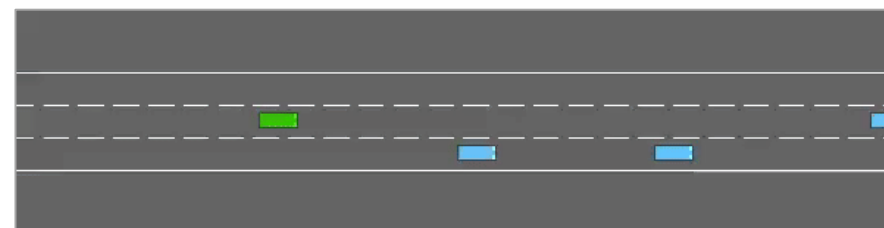


$$R_{\text{total}}(s, a) = \alpha R_{\text{safety}}(s, a) + \beta R_{\text{efficiency}}(s, a) + \gamma R_{\text{LLM}}(s, a)$$

An example of conservative model



An example of aggressive model



| Index        | Mean Reward    | Lane Change    | Speed Up       |
|--------------|----------------|----------------|----------------|
| DQN baseline | 0.82824        | 0.30681        | 0.42045        |
| Aggressive   | <b>0.83888</b> | 0.02326        | <b>0.83721</b> |
| Conservative | 0.71391        | <b>0.01333</b> | 0.00666        |
| Base         | 0.80140        | 0.10345        | 0.10345        |

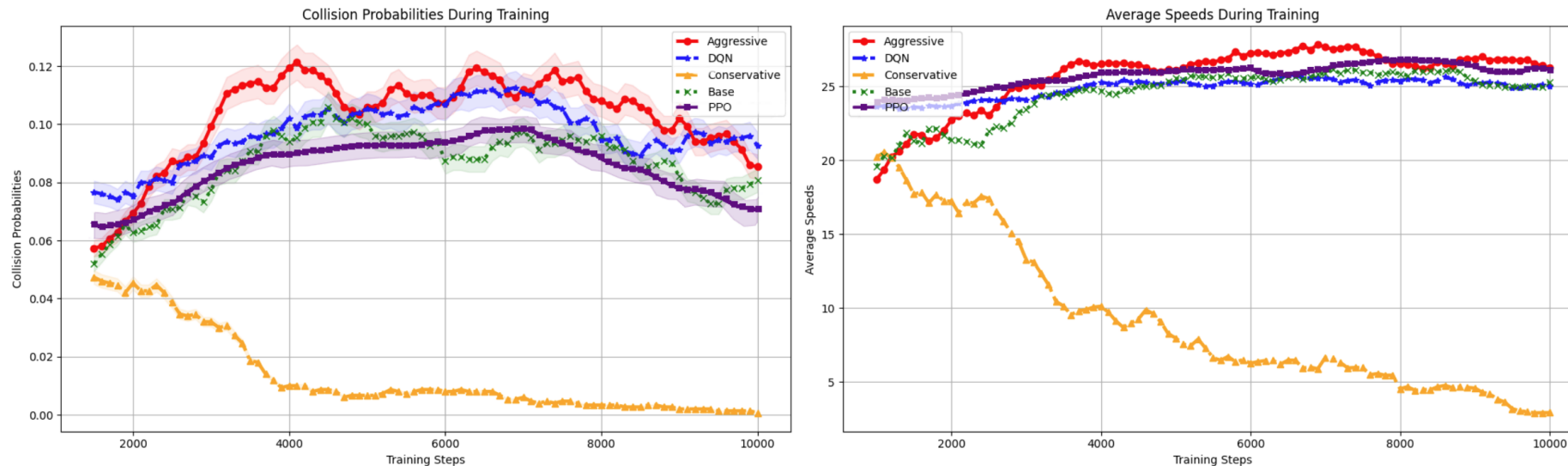


Fig. 6: Training results for driving style agents.

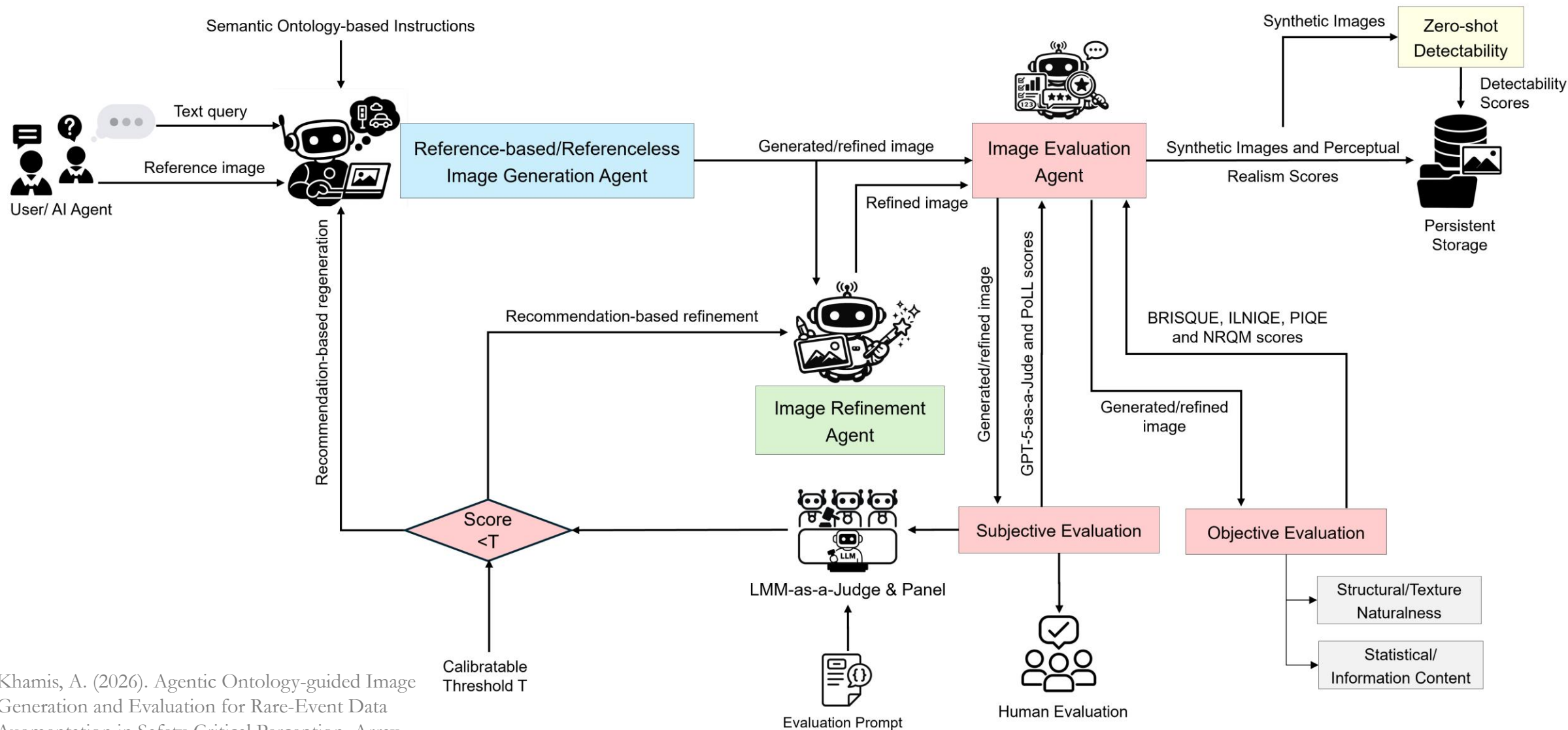
TABLE I: Experiment result: behavior analysis including PPO baseline.

| Index        | Mean Score     | Lane Change Score | Speed Up Score |
|--------------|----------------|-------------------|----------------|
| DQN baseline | 0.82824        | 0.30681           | 0.42045        |
| PPO baseline | 0.81000        | 0.20000           | 0.50000        |
| Aggressive   | 0.83888        | 0.02326           | 0.83721        |
| Conservative | 0.71391        | 0.01333           | 0.00666        |
| Base         | 0.80140        | 0.10345           | 0.10345        |
| BC-SAC [33]  | 0.83410        | 0.01750           | 0.75530        |
| LLM-RL       | <b>0.84532</b> | <b>0.01045</b>    | <b>0.81233</b> |

TABLE II: Reward ablation study: reward breakdown.

| Configuration       | Collision Score | Lane Change Score | High Speed Score |
|---------------------|-----------------|-------------------|------------------|
| Safety Only         | <b>-0.05</b>    | 0.23              | 0.18             |
| Efficiency Only     | -0.20           | <b>0.48</b>       | <b>0.72</b>      |
| LLM Only            | -0.10           | 0.31              | 0.33             |
| Safety + Efficiency | -0.12           | 0.35              | 0.55             |
| Safety + LLM        | -0.08           | 0.28              | 0.42             |
| Efficiency + LLM    | -0.15           | 0.41              | 0.63             |
| All (Full Reward)   | -0.12           | <b>0.53</b>       | 0.75             |

# Rare-Event Data Augmentation

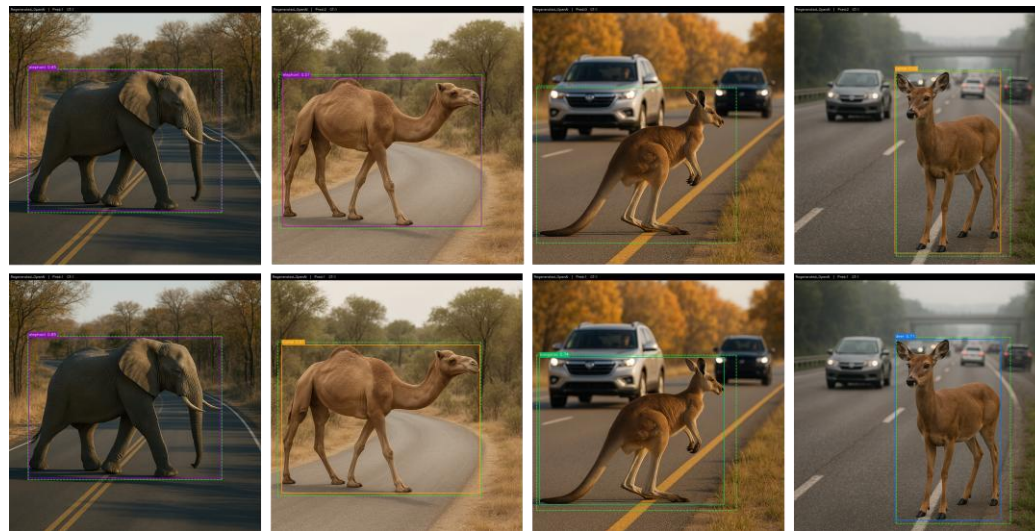


# Rare-Event Data Augmentation



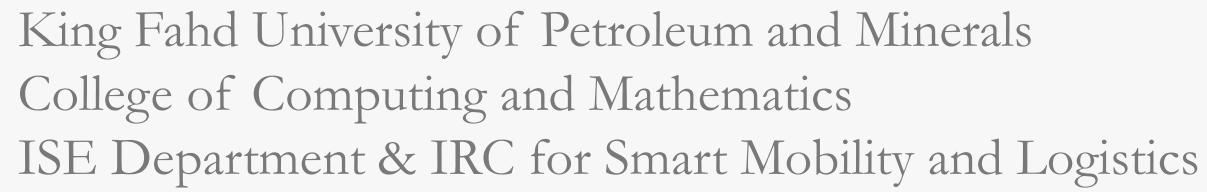
| Method                             | BRISQUE ↓ | ILNIQE ↓ | PIQE ↓ | NRQM ↑ |
|------------------------------------|-----------|----------|--------|--------|
| Referenceless OpenAI (Regenerated) | 19.47     | 48.76    | 34.10  | 7.39   |
| Referenceless Gemini 3 Pro         | 9.67      | 41.31    | 32.12  | 8.36   |
| Referenceless Grok 4.20            | 11.55     | 42.99    | 34.48  | 8.13   |

| Method                       | LMM-as-a-Judge ↑ | Panel of LMMs ↑ | Human Evaluation ↑ |
|------------------------------|------------------|-----------------|--------------------|
| OpenAI (Regenerated)         | 4.38             | 3.90            | 2.45               |
| Gemini 3 Pro (Referenceless) | 4.74             | 5.00            | 3.54               |
| Grok 4.20                    | 3.60             | 4.00            | 2.31               |



**Figure:** Examples of generated images using a referenceless ontology-guided approach. (1) GPT-5 synthesis, (b): gpt-image-1 refinement, (c) Recommendation-driven regeneration, (d) Gemini 3 Pro synthesis, (e) Grok 4.20

Khamis, A. (2026). Agentic Ontology-guided Image Generation and Evaluation for Rare-Event Data Augmentation in Safety-Critical Perception. Array. <https://doi.org/10.1016/j.array.2026.100932>. [GitHub](#)



April 16, 2026